A complete catalog of Hecke-algebra representations
for the graphs of `catalogue.pdf`

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For each vertex-transitive graph Γ in `catalogue.pdf` whose Hecke algebra is noncommutative, this catalog records:

- the *Sabidussi pair* (G, H): $G = \mathrm{Aut}\,\Gamma$ (or the chosen transitive subgroup) and $H = \mathrm{Stab}_G(v)$, so that Γ is a coset graph on G/H;
- the Wedderburn decomposition of $\mathcal{H} = \mathcal{H}(G, H) = \mathrm{End}_G\,\mathbb{C}[G/H]$;
- the table of *all* one-dimensional characters $T_i \mapsto \lambda_i$ (columns labelled by the G-irreducible χ they sit on, with its degree);
- for *every* irreducible representation of dimension $m \geq 2$:
  - a minimal generating set $A, B, \ldots$ chosen among the orbital images $\rho(T_i)$, as explicit matrices;
  - *complete* defining relations: the minimal polynomial of each generator, together with rewriting rules expressing $w \cdot g$ in the monomial basis for every basis word w and generator g — a presentation of $M_m(\mathbb{C})$ from which any word in the generators reduces;
  - the explicit matrix of *every* orbital generator $\rho(T_1), \ldots, \rho(T_r)$, each accompanied by its expansion as a linear combination of words in the block generators.

Throughout, $T_1 = I, T_2, \ldots, T_r$ is the orbital basis of $\mathcal{H}$, sorted by suborbit length k_i (ties among equal valencies are ordered by the orbit enumeration of the generating GAP session; the numbering is therefore fixed unambiguously by the explicit matrices below, but may differ from earlier write-ups within a tied valency class). All matrices were computed exactly over cyclotomic fields by isotypic restriction (`hecke_catalog_full.g`) and verified by reconstructing the m^2-dimensional matrix algebra, checking $\rho(\sum_i T_i) = 0$ on non-principal blocks, and re-multiplying every word expansion.

Notation. $\eta_a = \sum_{s \in H_p} \zeta_p^{as}$ denotes the sextic Gauss period of $\mathbb{Q}(\zeta_p)$ over the index-6 subgroup $H_p \leq (\mathbb{Z}/p)^\times$; $c_k = \zeta_7^k + \zeta_7^{-k}$ (so $c_1 + c_2 + c_3 = -1$); $\alpha = \frac{-1+\sqrt{-7}}{2}$ (so $\alpha\bar{\alpha} = 2$, $\alpha + \bar{\alpha} = -1$); $\omega = e^{2\pi i/3}$ (so $\omega\bar{\omega} = 1$, $\omega + \bar{\omega} = -1$).

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1 Dodecahedron, refined: $\big(\mathrm{PSL}(2,5),\, C_3\big)$

Graph. The dodecahedron on $20 = |\mathrm{PSL}(2,5)|/3$ vertices. Under the full group $C_2 \times A_5$ the pair is Gelfand; the representation theory below belongs to the refined pair $G = \mathrm{PSL}(2,5) \cong A_5$, $H = C_3$.

Algebra. Rank 8, $\mathcal{H} \cong \mathbb{C}^4 \oplus M_2(\mathbb{C})$.

Highlights. The valency-3 generator satisfies $B^2 + B + 3I = 0$ ($\det B = 3 =$ valency); the golden $\pm\sqrt{5}$ of the dodecahedron spectrum lives entirely in the 1-dimensional characters below. The "why-N" constant of this 20-point Meinhold miniature is 4: $(\rho(T_6) - I)^2 = 4I$.

One-dimensional characters.

	$\chi_1\,(1)$	$\chi_2\,(3)$	$\chi_3\,(3)$	$\chi_5\,(5)$
T_1	1	1	1	1
T_2	1	-1	-1	1
T_3	3	$-\sqrt{5}$	$\sqrt{5}$	1
T_4	3	$\sqrt{5}$	$-\sqrt{5}$	1
T_5	3	1	1	-1
T_6	3	-1	-1	-1
T_7	3	1	1	-1
T_8	3	-1	-1	-1

Block 1: $M_2(\mathbb{C})$ on χ_4 (degree 4)

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_5) = \begin{pmatrix} \frac{1}{2} & 1 \\ -\frac{15}{4} & -\frac{3}{2} \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$A^2 - I = 0,$$
$$B^2 + B + 3I = 0,$$
$$BA = 2I - A - AB,$$
$$ABA = -I + 2\,A - B,$$
$$ABB = -3\,A - AB.$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$

$$\rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A \qquad (k_2 = 1)$$

$$\rho(T_3) = \begin{pmatrix} -2 & 0 \\ 0 & 0 \end{pmatrix} = -I - A \qquad (k_3 = 3)$$

$$\rho(T_4) = \begin{pmatrix} -2 & 0 \\ 0 & 0 \end{pmatrix} = -I - A \qquad (k_4 = 3)$$

$$\rho(T_5) = \begin{pmatrix} \frac{1}{2} & 1 \\ -\frac{15}{4} & -\frac{3}{2} \end{pmatrix} = B \qquad (k_5 = 3)$$

$$\rho(T_6) = \begin{pmatrix} \frac{1}{2} & -1 \\ -\frac{15}{4} & \frac{3}{2} \end{pmatrix} = 2I - A - AB \qquad (k_6 = 3)$$

$$\rho(T_7) = \begin{pmatrix} \frac{1}{2} & -1 \\ \frac{15}{4} & -\frac{3}{2} \end{pmatrix} = -I + 2\,A - B \qquad (k_7 = 3)$$

$$\rho(T_8) = \begin{pmatrix} \frac{1}{2} & 1 \\ \frac{15}{4} & \frac{3}{2} \end{pmatrix} = AB \qquad (k_8 = 3).$$

2 F26A: $\left(C_{13}{:}C_6,\, C_3\right)$

Graph. F26A, the unique cubic arc-transitive graph on 26 vertices; $G = C_{13}{:}C_6$ ($|G| = 78$) is *sharply* arc-transitive, $H = C_3$.

Algebra. Rank 10, $\mathcal{H} \cong \mathbb{C}^2 \oplus M_2(\mathbb{C}) \oplus M_2(\mathbb{C})$, the two blocks Galois conjugate over $\mathbb{Q}(\sqrt{13})$ (the substitution $\sqrt{13} \mapsto -\sqrt{13}$ maps Block 1 to Block 2).

Highlights. With $\eta_0, \eta_1 = \frac{-1 \pm \sqrt{13}}{2}$ the sextic Gauss periods of $\mathbb{Q}(\zeta_{13})$, the traceless generators satisfy

$$B^2 = \frac{5 - \sqrt{13}}{2}\, I \quad \text{(Block 1)}, \qquad B^2 = \frac{5 + \sqrt{13}}{2}\, I \quad \text{(Block 2)}, \qquad N_+ N_- = 3 = \text{valency},$$

and $\frac{5 \mp \sqrt{13}}{2} = 3 + \eta$: *valency plus a Gauss period* (see the Remark after CR(31)).

One-dimensional characters.

	$\chi_1\,(1)$	$\chi_2\,(1)$
T_1	1	1
T_2	1	-1
T_3	3	-3
T_4	3	3
T_5	3	-3
T_6	3	3
T_7	3	-3
T_8	3	3
T_9	3	-3
T_{10}	3	3

Block 1: $M_2(\mathbb{C})$ **on** χ_7 **(degree 6)**

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} -\frac{1}{4} + \frac{1}{4}\sqrt{13} & 1 \\ \frac{13}{8} - \frac{3}{8}\sqrt{13} & \frac{1}{4} - \frac{1}{4}\sqrt{13} \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$\begin{gathered} A^2 - I = 0, \\ B^2 + \left(-\tfrac{5}{2} + \tfrac{1}{2}\sqrt{13}\right) I = 0, \\ BA = \left(-\tfrac{1}{2} + \tfrac{1}{2}\sqrt{13}\right) I - AB, \\ ABA = \left(-\tfrac{1}{2} + \tfrac{1}{2}\sqrt{13}\right) A - B, \\ ABB = \left(\tfrac{5}{2} - \tfrac{1}{2}\sqrt{13}\right) A. \end{gathered}$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\begin{aligned}
\rho(T_1) &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I && (k_1 = 1) \\
\rho(T_2) &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A && (k_2 = 1) \\
\rho(T_3) &= \begin{pmatrix} -\frac{1}{4} + \frac{1}{4}\sqrt{13} & 1 \\ \frac{13}{8} - \frac{3}{8}\sqrt{13} & \frac{1}{4} - \frac{1}{4}\sqrt{13} \end{pmatrix} = B && (k_3 = 3) \\
\rho(T_4) &= \begin{pmatrix} -\frac{1}{4} - \frac{1}{4}\sqrt{13} & \frac{3}{2} + \frac{1}{2}\sqrt{13} \\ -\frac{1}{4}\sqrt{13} & -\frac{1}{4} - \frac{1}{4}\sqrt{13} \end{pmatrix} && (k_4 = 3) \\
&= \left(-\tfrac{3}{2} - \tfrac{1}{2}\sqrt{13}\right) I + \left(\tfrac{3}{2} + \tfrac{1}{2}\sqrt{13}\right) AB \\
\rho(T_5) &= \begin{pmatrix} -\frac{1}{4} + \frac{1}{4}\sqrt{13} & -1 \\ -\frac{13}{8} + \frac{3}{8}\sqrt{13} & \frac{1}{4} - \frac{1}{4}\sqrt{13} \end{pmatrix} = \left(-\tfrac{1}{2} + \tfrac{1}{2}\sqrt{13}\right) A - B && (k_5 = 3) \\
\rho(T_6) &= \begin{pmatrix} -\frac{1}{4} + \frac{1}{4}\sqrt{13} & -1 \\ \frac{13}{8} - \frac{3}{8}\sqrt{13} & -\frac{1}{4} + \frac{1}{4}\sqrt{13} \end{pmatrix} = \left(-\tfrac{1}{2} + \tfrac{1}{2}\sqrt{13}\right) I - AB && (k_6 = 3) \\
\rho(T_7) &= \begin{pmatrix} -\frac{1}{4} - \frac{1}{4}\sqrt{13} & -\frac{3}{2} - \frac{1}{2}\sqrt{13} \\ -\frac{1}{4}\sqrt{13} & \frac{1}{4} + \frac{1}{4}\sqrt{13} \end{pmatrix} = A + \left(-\tfrac{3}{2} - \tfrac{1}{2}\sqrt{13}\right) B && (k_7 = 3) \\
\rho(T_8) &= \begin{pmatrix} -\frac{1}{4} - \frac{1}{4}\sqrt{13} & -\frac{3}{2} - \frac{1}{2}\sqrt{13} \\ \frac{1}{4}\sqrt{13} & -\frac{1}{4} - \frac{1}{4}\sqrt{13} \end{pmatrix} = I + \left(-\tfrac{3}{2} - \tfrac{1}{2}\sqrt{13}\right) AB && (k_8 = 3) \\
\rho(T_9) &= \begin{pmatrix} -\frac{1}{4} - \frac{1}{4}\sqrt{13} & \frac{3}{2} + \frac{1}{2}\sqrt{13} \\ \frac{1}{4}\sqrt{13} & \frac{1}{4} + \frac{1}{4}\sqrt{13} \end{pmatrix} && (k_9 = 3) \\
&= \left(-\tfrac{3}{2} - \tfrac{1}{2}\sqrt{13}\right) A + \left(\tfrac{3}{2} + \tfrac{1}{2}\sqrt{13}\right) B \\
\rho(T_{10}) &= \begin{pmatrix} -\frac{1}{4} + \frac{1}{4}\sqrt{13} & 1 \\ -\frac{13}{8} + \frac{3}{8}\sqrt{13} & -\frac{1}{4} + \frac{1}{4}\sqrt{13} \end{pmatrix} = AB && (k_{10} = 3).
\end{aligned}$$

Block 2: $M_2(\mathbb{C})$ **on** χ_8 **(degree 6)**

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} -\frac{1}{4} - \frac{1}{4}\sqrt{13} & 1 \\ \frac{13}{8} + \frac{3}{8}\sqrt{13} & \frac{1}{4} + \frac{1}{4}\sqrt{13} \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$\begin{gathered}A^2 - I = 0,\\ B^2 + \left(-\tfrac{5}{2} - \tfrac{1}{2}\sqrt{13}\right)I = 0,\\ BA = \left(-\tfrac{1}{2} - \tfrac{1}{2}\sqrt{13}\right)I - AB,\\ ABA = \left(-\tfrac{1}{2} - \tfrac{1}{2}\sqrt{13}\right)A - B,\\ ABB = \left(\tfrac{5}{2} + \tfrac{1}{2}\sqrt{13}\right)A.\end{gathered}$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\begin{aligned}
\rho(T_1) &= \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} = I && (k_1 = 1)\\
\rho(T_2) &= \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} = A && (k_2 = 1)\\
\rho(T_3) &= \begin{pmatrix}-\frac{1}{4} - \frac{1}{4}\sqrt{13} & 1\\ \frac{13}{8} + \frac{3}{8}\sqrt{13} & \frac{1}{4} + \frac{1}{4}\sqrt{13}\end{pmatrix} = B && (k_3 = 3)\\
\rho(T_4) &= \begin{pmatrix}-\frac{1}{4} + \frac{1}{4}\sqrt{13} & \frac{3}{2} - \frac{1}{2}\sqrt{13}\\ \frac{1}{4}\sqrt{13} & -\frac{1}{4} + \frac{1}{4}\sqrt{13}\end{pmatrix} && (k_4 = 3)\\
&= \left(-\tfrac{3}{2} + \tfrac{1}{2}\sqrt{13}\right)I + \left(\tfrac{3}{2} - \tfrac{1}{2}\sqrt{13}\right)AB\\
\rho(T_5) &= \begin{pmatrix}-\frac{1}{4} - \frac{1}{4}\sqrt{13} & -1\\ -\frac{13}{8} - \frac{3}{8}\sqrt{13} & \frac{1}{4} + \frac{1}{4}\sqrt{13}\end{pmatrix} = \left(-\tfrac{1}{2} - \tfrac{1}{2}\sqrt{13}\right)A - B && (k_5 = 3)\\
\rho(T_6) &= \begin{pmatrix}-\frac{1}{4} - \frac{1}{4}\sqrt{13} & -1\\ \frac{13}{8} + \frac{3}{8}\sqrt{13} & -\frac{1}{4} - \frac{1}{4}\sqrt{13}\end{pmatrix} = \left(-\tfrac{1}{2} - \tfrac{1}{2}\sqrt{13}\right)I - AB && (k_6 = 3)\\
\rho(T_7) &= \begin{pmatrix}-\frac{1}{4} + \frac{1}{4}\sqrt{13} & -\frac{3}{2} + \frac{1}{2}\sqrt{13}\\ \frac{1}{4}\sqrt{13} & \frac{1}{4} - \frac{1}{4}\sqrt{13}\end{pmatrix} = A + \left(-\tfrac{3}{2} + \tfrac{1}{2}\sqrt{13}\right)B && (k_7 = 3)\\
\rho(T_8) &= \begin{pmatrix}-\frac{1}{4} + \frac{1}{4}\sqrt{13} & -\frac{3}{2} + \frac{1}{2}\sqrt{13}\\ -\frac{1}{4}\sqrt{13} & -\frac{1}{4} + \frac{1}{4}\sqrt{13}\end{pmatrix} = I + \left(-\tfrac{3}{2} + \tfrac{1}{2}\sqrt{13}\right)AB && (k_8 = 3)\\
\rho(T_9) &= \begin{pmatrix}-\frac{1}{4} + \frac{1}{4}\sqrt{13} & \frac{3}{2} - \frac{1}{2}\sqrt{13}\\ -\frac{1}{4}\sqrt{13} & \frac{1}{4} - \frac{1}{4}\sqrt{13}\end{pmatrix} && (k_9 = 3)\\
&= \left(-\tfrac{3}{2} + \tfrac{1}{2}\sqrt{13}\right)A + \left(\tfrac{3}{2} - \tfrac{1}{2}\sqrt{13}\right)B\\
\rho(T_{10}) &= \begin{pmatrix}-\frac{1}{4} - \frac{1}{4}\sqrt{13} & 1\\ -\frac{13}{8} - \frac{3}{8}\sqrt{13} & -\frac{1}{4} - \frac{1}{4}\sqrt{13}\end{pmatrix} = AB && (k_{10} = 3).
\end{aligned}$$

3 CR(19): $\left(C_{19}{:}C_6,\ C_3\right)$

Graph. The 38-vertex cubic arc-transitive Frobenius graph; $G = C_{19}{:}C_6$ ($|G| = 114$), $H = C_3$. Here $H_{19} = \langle 8\rangle = \{1, 7, 8, 11, 12, 18\} \le (\mathbb{Z}/19)^\times$ and the three sextic periods are η_1, η_2, η_4 (indexed by coset representatives $1, 2, 4$), satisfying $\eta_1 + \eta_2 + \eta_4 = -1$; entries below are written in the basis $\{1, \eta_a, \eta_b\}$ that gives the sparsest expression.

Algebra. Rank 14, $\mathcal{H} \cong \mathbb{C}^2 \oplus M_2(\mathbb{C})^{\oplus 3}$.

Highlights. The three blocks are permuted cyclically by the Galois action $a \mapsto 2a$ on period indices; in each, the traceless valency-3 orbital squares to $(3 + \eta)I$.

One-dimensional characters.

	$\chi_1\,(1)$	$\chi_2\,(1)$
T_1	1	1
T_2	1	-1
T_3	3	3
T_4	3	-3
T_5	3	3
T_6	3	-3
T_7	3	-3
T_8	3	3
T_9	3	-3
T_{10}	3	3
T_{11}	3	-3
T_{12}	3	3
T_{13}	3	-3
T_{14}	3	3

Block 1: $M_2(\mathbb{C})$ **on** χ_7 **(degree 6)**

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} \frac{1}{2}\eta_4 & 1 \\ -2 - \frac{1}{4}\eta_1 - \eta_2 & \frac{1}{2}\eta_4 \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$\begin{gathered} A^2 - I = 0, \\ B^2 - \eta_4\, B + \big(3 + \eta_2\big) I = 0, \\ BA = \eta_4\, A - AB, \\ ABA = \eta_4 I - B, \\ ABB = \big(-3 - \eta_2\big) A + \eta_4\, AB. \end{gathered}$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$

$$\rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A \qquad (k_2 = 1)$$

$$\rho(T_3) = \begin{pmatrix} \frac{1}{2}\eta_4 & 1 \\ -2 - \frac{1}{4}\eta_1 - \eta_2 & \frac{1}{2}\eta_4 \end{pmatrix} = B \qquad (k_3 = 3)$$

$$\rho(T_4) = \begin{pmatrix} \frac{1}{2}\eta_4 & -1 \\ -2 - \frac{1}{4}\eta_1 - \eta_2 & -\frac{1}{2}\eta_4 \end{pmatrix} = \eta_4\, A - AB \qquad (k_4 = 3)$$

$$\rho(T_5) = \begin{pmatrix} \frac{1}{2}\eta_1 & 2 + \eta_4 \\ \frac{1}{4}\eta_2 - \frac{1}{4}\eta_4 & \frac{1}{2}\eta_1 \end{pmatrix} = \big(-1 + 2\eta_1 + \eta_2\big) I + \big(2 + \eta_4\big) B \qquad (k_5 = 3)$$

$$\rho(T_6) = \begin{pmatrix} \frac{1}{2}\eta_1 & -2 - \eta_4 \\ \frac{1}{4}\eta_2 - \frac{1}{4}\eta_4 & -\frac{1}{2}\eta_1 \end{pmatrix} = \big(2 + \eta_4\big) A + \big(-2 - \eta_4\big) AB \qquad (k_6 = 3)$$

$$\rho(T_7) = \begin{pmatrix} \frac{1}{2}\eta_4 & 1 \\ 2 + \frac{1}{4}\eta_1 + \eta_2 & -\frac{1}{2}\eta_4 \end{pmatrix} = AB \qquad (k_7 = 3)$$

$$\rho(T_8) = \begin{pmatrix} \frac{1}{2}\eta_4 & -1 \\ 2 + \frac{1}{4}\eta_1 + \eta_2 & \frac{1}{2}\eta_4 \end{pmatrix} = \eta_4 I - B \qquad (k_8 = 3)$$

$$\rho(T_9) = \begin{pmatrix} \frac{1}{2}\eta_1 & 2 + \eta_4 \\ -\frac{1}{4}\eta_2 + \frac{1}{4}\eta_4 & -\frac{1}{2}\eta_1 \end{pmatrix} \qquad (k_9 = 3)$$

$$= \big(-1 + 2\eta_1 + \eta_2\big)A + \big(2 + \eta_4\big)AB$$

$$\rho(T_{10}) = \begin{pmatrix} \frac{1}{2}\eta_1 & -2 - \eta_4 \\ -\frac{1}{4}\eta_2 + \frac{1}{4}\eta_4 & \frac{1}{2}\eta_1 \end{pmatrix} = \big(2 + \eta_4\big)I + \big(-2 - \eta_4\big)B \qquad (k_{10} = 3)$$

$$\rho(T_{11}) = \begin{pmatrix} \frac{1}{2}\eta_2 & -1 - \eta_1 \\ -\frac{1}{4}\eta_1 + \frac{1}{4}\eta_2 & -\frac{1}{2}\eta_2 \end{pmatrix} = \big(-1 + \eta_2\big)A + \big(-1 - \eta_1\big)AB \qquad (k_{11} = 3)$$

$$\rho(T_{12}) = \begin{pmatrix} \frac{1}{2}\eta_2 & 1 + \eta_1 \\ -\frac{1}{4}\eta_1 + \frac{1}{4}\eta_2 & \frac{1}{2}\eta_2 \end{pmatrix} = I + \big(1 + \eta_1\big)B \qquad (k_{12} = 3)$$

$$\rho(T_{13}) = \begin{pmatrix} \frac{1}{2}\eta_2 & 1 + \eta_1 \\ \frac{1}{4}\eta_1 - \frac{1}{4}\eta_2 & -\frac{1}{2}\eta_2 \end{pmatrix} = A + \big(1 + \eta_1\big)AB \qquad (k_{13} = 3)$$

$$\rho(T_{14}) = \begin{pmatrix} \frac{1}{2}\eta_2 & -1 - \eta_1 \\ \frac{1}{4}\eta_1 - \frac{1}{4}\eta_2 & \frac{1}{2}\eta_2 \end{pmatrix} = \big(-1 + \eta_2\big)I + \big(-1 - \eta_1\big)B \qquad (k_{14} = 3).$$

Block 2: $M_2(\mathbb{C})$ **on** χ_8 **(degree 6)**

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} \frac{1}{2}\eta_1 & 1 \\ -1 + \eta_1 + \frac{3}{4}\eta_2 & \frac{1}{2}\eta_1 \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$A^2 - I = 0,$$
$$B^2 - \eta_1\, B + \big(3 + \eta_4\big)I = 0,$$
$$BA = \eta_1\, A - AB,$$
$$ABA = \eta_1 I - B,$$
$$ABB = \big(-3 - \eta_4\big)A + \eta_1\, AB.$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$

$$\rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A \qquad (k_2 = 1)$$

$$\rho(T_3) = \begin{pmatrix} \frac{1}{2}\eta_1 & 1 \\ -1 + \eta_1 + \frac{3}{4}\eta_2 & \frac{1}{2}\eta_1 \end{pmatrix} = B \qquad (k_3 = 3)$$

$$\rho(T_4) = \begin{pmatrix} \frac{1}{2}\eta_1 & -1 \\ -1 + \eta_1 + \frac{3}{4}\eta_2 & -\frac{1}{2}\eta_1 \end{pmatrix} = \eta_1\, A - AB \qquad (k_4 = 3)$$

$$\rho(T_5) = \begin{pmatrix} \frac{1}{2}\eta_2 & 2 + \eta_1 \\ -\frac{1}{4}\eta_1 + \frac{1}{4}\eta_4 & \frac{1}{2}\eta_2 \end{pmatrix} = \big(-2 - \eta_1 + \eta_2\big)I + \big(2 + \eta_1\big)B \qquad (k_5 = 3)$$

$$\rho(T_6) = \begin{pmatrix} \frac{1}{2}\eta_2 & -2 - \eta_1 \\ -\frac{1}{4}\eta_1 + \frac{1}{4}\eta_4 & -\frac{1}{2}\eta_2 \end{pmatrix} = \big(2 + \eta_1\big)A + \big(-2 - \eta_1\big)AB \qquad (k_6 = 3)$$

$$\rho(T_7) = \begin{pmatrix} \frac{1}{2}\eta_1 & 1 \\ 1-\eta_1-\frac{3}{4}\eta_2 & -\frac{1}{2}\eta_1 \end{pmatrix} = AB \qquad (k_7 = 3)$$

$$\rho(T_8) = \begin{pmatrix} \frac{1}{2}\eta_1 & -1 \\ 1-\eta_1-\frac{3}{4}\eta_2 & \frac{1}{2}\eta_1 \end{pmatrix} = \eta_1 I - B \qquad (k_8 = 3)$$

$$\rho(T_9) = \begin{pmatrix} \frac{1}{2}\eta_2 & 2+\eta_1 \\ \frac{1}{4}\eta_1-\frac{1}{4}\eta_4 & -\frac{1}{2}\eta_2 \end{pmatrix} = \big(-2-\eta_1+\eta_2\big)A + \big(2+\eta_1\big)AB \qquad (k_9 = 3)$$

$$\rho(T_{10}) = \begin{pmatrix} \frac{1}{2}\eta_2 & -2-\eta_1 \\ \frac{1}{4}\eta_1-\frac{1}{4}\eta_4 & \frac{1}{2}\eta_2 \end{pmatrix} = \big(2+\eta_1\big)I + \big(-2-\eta_1\big)B \qquad (k_{10} = 3)$$

$$\rho(T_{11}) = \begin{pmatrix} \frac{1}{2}\eta_4 & -1-\eta_2 \\ -\frac{1}{4}\eta_2+\frac{1}{4}\eta_4 & -\frac{1}{2}\eta_4 \end{pmatrix} = \big(-1+\eta_4\big)A + \big(-1-\eta_2\big)AB \qquad (k_{11} = 3)$$

$$\rho(T_{12}) = \begin{pmatrix} \frac{1}{2}\eta_4 & 1+\eta_2 \\ -\frac{1}{4}\eta_2+\frac{1}{4}\eta_4 & \frac{1}{2}\eta_4 \end{pmatrix} = I + \big(1+\eta_2\big)B \qquad (k_{12} = 3)$$

$$\rho(T_{13}) = \begin{pmatrix} \frac{1}{2}\eta_4 & 1+\eta_2 \\ \frac{1}{4}\eta_2-\frac{1}{4}\eta_4 & -\frac{1}{2}\eta_4 \end{pmatrix} = A + \big(1+\eta_2\big)AB \qquad (k_{13} = 3)$$

$$\rho(T_{14}) = \begin{pmatrix} \frac{1}{2}\eta_4 & -1-\eta_2 \\ \frac{1}{4}\eta_2-\frac{1}{4}\eta_4 & \frac{1}{2}\eta_4 \end{pmatrix} = \big(-1+\eta_4\big)I + \big(-1-\eta_2\big)B \qquad (k_{14} = 3).$$

Block 3: $M_2(\mathbb{C})$ on χ_9 (degree 6)

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} \frac{1}{2}\eta_2 & 1 \\ -\frac{7}{4}-\frac{3}{4}\eta_1+\frac{1}{4}\eta_2 & \frac{1}{2}\eta_2 \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$\begin{gathered} A^2 - I = 0, \\ B^2 - \eta_2\, B + \big(3+\eta_1\big)I = 0, \\ BA = \eta_2\, A - AB, \\ ABA = \eta_2 I - B, \\ ABB = \big(-3-\eta_1\big)A + \eta_2\, AB. \end{gathered}$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$

$$\rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A \qquad (k_2 = 1)$$

$$\rho(T_3) = \begin{pmatrix} \frac{1}{2}\eta_2 & 1 \\ -\frac{7}{4}-\frac{3}{4}\eta_1+\frac{1}{4}\eta_2 & \frac{1}{2}\eta_2 \end{pmatrix} = B \qquad (k_3 = 3)$$

$$\rho(T_4) = \begin{pmatrix} \frac{1}{2}\eta_2 & -1 \\ -\frac{7}{4}-\frac{3}{4}\eta_1+\frac{1}{4}\eta_2 & -\frac{1}{2}\eta_2 \end{pmatrix} = \eta_2\, A - AB \qquad (k_4 = 3)$$

$$\rho(T_5) = \begin{pmatrix} \frac{1}{2}\eta_4 & 2+\eta_2 \\ \frac{1}{4}\eta_1-\frac{1}{4}\eta_2 & \frac{1}{2}\eta_4 \end{pmatrix} = \big(-3-\eta_1-2\eta_2\big)I + \big(2+\eta_2\big)B \qquad (k_5 = 3)$$

$$\rho(T_6) = \begin{pmatrix} \frac{1}{2}\eta_4 & -2-\eta_2 \\ \frac{1}{4}\eta_1-\frac{1}{4}\eta_2 & -\frac{1}{2}\eta_4 \end{pmatrix} = \big(2+\eta_2\big)A + \big(-2-\eta_2\big)AB \qquad (k_6 = 3)$$

$$\rho(T_7) = \begin{pmatrix} \frac{1}{2}\eta_2 & 1 \\ \frac{7}{4} + \frac{3}{4}\eta_1 - \frac{1}{4}\eta_2 & -\frac{1}{2}\eta_2 \end{pmatrix} = AB \qquad (k_7 = 3)$$

$$\rho(T_8) = \begin{pmatrix} \frac{1}{2}\eta_2 & -1 \\ \frac{7}{4} + \frac{3}{4}\eta_1 - \frac{1}{4}\eta_2 & \frac{1}{2}\eta_2 \end{pmatrix} = \eta_2 I - B \qquad (k_8 = 3)$$

$$\rho(T_9) = \begin{pmatrix} \frac{1}{2}\eta_4 & 2 + \eta_2 \\ -\frac{1}{4}\eta_1 + \frac{1}{4}\eta_2 & -\frac{1}{2}\eta_4 \end{pmatrix} \qquad (k_9 = 3)$$

$$= \big(-3 - \eta_1 - 2\eta_2\big)A + \big(2 + \eta_2\big)AB$$

$$\rho(T_{10}) = \begin{pmatrix} \frac{1}{2}\eta_4 & -2 - \eta_2 \\ -\frac{1}{4}\eta_1 + \frac{1}{4}\eta_2 & \frac{1}{2}\eta_4 \end{pmatrix} = \big(2 + \eta_2\big)I + \big(-2 - \eta_2\big)B \qquad (k_{10} = 3)$$

$$\rho(T_{11}) = \begin{pmatrix} \frac{1}{2}\eta_1 & \eta_1 + \eta_2 \\ \frac{1}{4}\eta_1 - \frac{1}{4}\eta_4 & -\frac{1}{2}\eta_1 \end{pmatrix} = \big(-1 + \eta_1\big)A + \big(\eta_1 + \eta_2\big)AB \qquad (k_{11} = 3)$$

$$\rho(T_{12}) = \begin{pmatrix} \frac{1}{2}\eta_1 & -\eta_1 - \eta_2 \\ \frac{1}{4}\eta_1 - \frac{1}{4}\eta_4 & \frac{1}{2}\eta_1 \end{pmatrix} = I + \big(-\eta_1 - \eta_2\big)B \qquad (k_{12} = 3)$$

$$\rho(T_{13}) = \begin{pmatrix} \frac{1}{2}\eta_1 & -\eta_1 - \eta_2 \\ -\frac{1}{4}\eta_1 + \frac{1}{4}\eta_4 & -\frac{1}{2}\eta_1 \end{pmatrix} = A + \big(-\eta_1 - \eta_2\big)AB \qquad (k_{13} = 3)$$

$$\rho(T_{14}) = \begin{pmatrix} \frac{1}{2}\eta_1 & \eta_1 + \eta_2 \\ -\frac{1}{4}\eta_1 + \frac{1}{4}\eta_4 & \frac{1}{2}\eta_1 \end{pmatrix} = \big(-1 + \eta_1\big)I + \big(\eta_1 + \eta_2\big)B \qquad (k_{14} = 3).$$

4 CR(31): $\big(C_{31}{:}C_6,\, C_3\big)$

Graph. The 62-vertex sibling; $G = C_{31}{:}C_6$ ($|G| = 186$), $H = C_3$. Here $H_{31} = \langle 26 \rangle = \{1, 5, 6, 25, 26, 30\}$, and there are five sextic periods $\eta_1, \eta_2, \eta_3, \eta_4, \eta_8$ (coset representatives $1, 2, 3, 4, 8$), summing to -1.

Algebra. Rank 22, $\mathcal{H} \cong \mathbb{C}^2 \oplus M_2(\mathbb{C})^{\oplus 5}$.

Highlights. Five Galois-conjugate blocks, cycled by $a \mapsto 2a$; same $(3 + \eta)$-law as CR(19).

One-dimensional characters.

	$\chi_1(1)$	$\chi_2(1)$
T_1	1	1
T_2	1	-1
T_3	3	3
T_4	3	-3
T_5	3	3
T_6	3	-3
T_7	3	-3
T_8	3	3
T_9	3	3
T_{10}	3	-3
T_{11}	3	-3
T_{12}	3	3
T_{13}	3	-3
T_{14}	3	3
T_{15}	3	3
T_{16}	3	-3
T_{17}	3	-3
T_{18}	3	3
T_{19}	3	-3
T_{20}	3	3
T_{21}	3	3
T_{22}	3	-3

Block 1: $M_2(\mathbb{C})$ on χ_7 (degree 6)

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} \frac{1}{2}\eta_3 & 1 \\ -\frac{3}{2} + \frac{1}{4}\eta_1 - \frac{1}{2}\eta_2 + \frac{1}{2}\eta_3 & \frac{1}{2}\eta_3 \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$\begin{gathered} A^2 - I = 0, \\ B^2 - \eta_3\, B + \big(3 + \eta_2\big) I = 0, \\ BA = \eta_3\, A - AB, \\ ABA = \eta_3 I - B, \\ ABB = \big(-3 - \eta_2\big) A + \eta_3\, AB. \end{gathered}$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$

$$\rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A \qquad (k_2 = 1)$$

$$\rho(T_3) = \begin{pmatrix} \frac{1}{2}\eta_3 & 1 \\ -\frac{3}{2} + \frac{1}{4}\eta_1 - \frac{1}{2}\eta_2 + \frac{1}{2}\eta_3 & \frac{1}{2}\eta_3 \end{pmatrix} = B \qquad (k_3 = 3)$$

$$\rho(T_4) = \begin{pmatrix} \frac{1}{2}\eta_3 & -1 \\ -\frac{3}{2} + \frac{1}{4}\eta_1 - \frac{1}{2}\eta_2 + \frac{1}{2}\eta_3 & -\frac{1}{2}\eta_3 \end{pmatrix} = \eta_3\, A - AB \qquad (k_4 = 3)$$

$$\rho(T_5) = \begin{pmatrix} \frac{1}{2}\eta_1 & 2+\eta_3 \\ \frac{1}{2}\eta_2 - \frac{1}{4}\eta_3 - \frac{1}{4}\eta_4 & \frac{1}{2}\eta_1 \end{pmatrix} \qquad (k_5 = 3)$$
$$= \big(-3 - \eta_2 - 2\eta_3\big)I + \big(2+\eta_3\big)B$$

$$\rho(T_6) = \begin{pmatrix} \frac{1}{2}\eta_1 & -2-\eta_3 \\ \frac{1}{2}\eta_2 - \frac{1}{4}\eta_3 - \frac{1}{4}\eta_4 & -\frac{1}{2}\eta_1 \end{pmatrix} \qquad (k_6 = 3)$$
$$= \big(3 + \eta_1 + \eta_2 + 2\eta_3\big)A + \big(-2-\eta_3\big)AB$$

$$\rho(T_7) = \begin{pmatrix} \frac{1}{2}\eta_3 & 1 \\ \frac{3}{2} - \frac{1}{4}\eta_1 + \frac{1}{2}\eta_2 - \frac{1}{2}\eta_3 & -\frac{1}{2}\eta_3 \end{pmatrix} = AB \qquad (k_7 = 3)$$

$$\rho(T_8) = \begin{pmatrix} \frac{1}{2}\eta_3 & -1 \\ \frac{3}{2} - \frac{1}{4}\eta_1 + \frac{1}{2}\eta_2 - \frac{1}{2}\eta_3 & \frac{1}{2}\eta_3 \end{pmatrix} = \eta_3 I - B \qquad (k_8 = 3)$$

$$\rho(T_9) = \begin{pmatrix} \frac{1}{2}\eta_8 & 3+\eta_1+\eta_2+2\eta_3 \\ -\frac{1}{2}\eta_1 + \frac{1}{4}\eta_2 + \frac{1}{4}\eta_8 & \frac{1}{2}\eta_8 \end{pmatrix} \qquad (k_9 = 3)$$
$$= \big(-5 - \eta_1 - 2\eta_2 - 4\eta_3\big)I + \big(3+\eta_1+\eta_2+2\eta_3\big)B$$

$$\rho(T_{10}) = \begin{pmatrix} \frac{1}{2}\eta_8 & -3-\eta_1-\eta_2-2\eta_3 \\ -\frac{1}{2}\eta_1 + \frac{1}{4}\eta_2 + \frac{1}{4}\eta_8 & -\frac{1}{2}\eta_8 \end{pmatrix} \qquad (k_{10} = 3)$$
$$= \big(4 + \eta_2 + 3\eta_3 - \eta_4\big)A + \big(-3-\eta_1-\eta_2-2\eta_3\big)AB$$

$$\rho(T_{11}) = \begin{pmatrix} \frac{1}{2}\eta_1 & 2+\eta_3 \\ -\frac{1}{2}\eta_2 + \frac{1}{4}\eta_3 + \frac{1}{4}\eta_4 & -\frac{1}{2}\eta_1 \end{pmatrix} \qquad (k_{11} = 3)$$
$$= \big(-3 - \eta_2 - 2\eta_3\big)A + \big(2+\eta_3\big)AB$$

$$\rho(T_{12}) = \begin{pmatrix} \frac{1}{2}\eta_1 & -2-\eta_3 \\ -\frac{1}{2}\eta_2 + \frac{1}{4}\eta_3 + \frac{1}{4}\eta_4 & \frac{1}{2}\eta_1 \end{pmatrix} \qquad (k_{12} = 3)$$
$$= \big(3 + \eta_1 + \eta_2 + 2\eta_3\big)I + \big(-2-\eta_3\big)B$$

$$\rho(T_{13}) = \begin{pmatrix} \frac{1}{2}\eta_2 & 2+\eta_3-\eta_4 \\ -\frac{1}{4}\eta_4 + \frac{1}{4}\eta_8 & -\frac{1}{2}\eta_2 \end{pmatrix} \qquad (k_{13} = 3)$$
$$= \big(-4 - \eta_1 - \eta_2 - 3\eta_3\big)A + \big(2+\eta_3-\eta_4\big)AB$$

$$\rho(T_{14}) = \begin{pmatrix} \frac{1}{2}\eta_2 & -2-\eta_3+\eta_4 \\ -\frac{1}{4}\eta_4 + \frac{1}{4}\eta_8 & \frac{1}{2}\eta_2 \end{pmatrix} \qquad (k_{14} = 3)$$
$$= \big(4 + \eta_1 + 2\eta_2 + 3\eta_3\big)I + \big(-2-\eta_3+\eta_4\big)B$$

$$\rho(T_{15}) = \begin{pmatrix} \frac{1}{2}\eta_2 & 2+\eta_3-\eta_4 \\ \frac{1}{4}\eta_4 - \frac{1}{4}\eta_8 & \frac{1}{2}\eta_2 \end{pmatrix} \qquad (k_{15} = 3)$$
$$= \big(-4 - \eta_1 - \eta_2 - 3\eta_3\big)I + \big(2+\eta_3-\eta_4\big)B$$

$$\rho(T_{16}) = \begin{pmatrix} \frac{1}{2}\eta_2 & -2-\eta_3+\eta_4 \\ \frac{1}{4}\eta_4 - \frac{1}{4}\eta_8 & -\frac{1}{2}\eta_2 \end{pmatrix} \qquad (k_{16} = 3)$$
$$= \big(4 + \eta_1 + 2\eta_2 + 3\eta_3\big)A + \big(-2-\eta_3+\eta_4\big)AB$$

$$\rho(T_{17}) = \begin{pmatrix} \frac{1}{2}\eta_8 & 3+\eta_1+\eta_2+2\eta_3 \\ \frac{1}{2}\eta_1 - \frac{1}{4}\eta_2 - \frac{1}{4}\eta_8 & -\frac{1}{2}\eta_8 \end{pmatrix} \qquad (k_{17} = 3)$$
$$= \big(-5 - \eta_1 - 2\eta_2 - 4\eta_3\big)A + \big(3+\eta_1+\eta_2+2\eta_3\big)AB$$

$$\rho(T_{18}) = \begin{pmatrix} \frac{1}{2}\eta_8 & -3-\eta_1-\eta_2-2\eta_3 \\ \frac{1}{2}\eta_1 - \frac{1}{4}\eta_2 - \frac{1}{4}\eta_8 & \frac{1}{2}\eta_8 \end{pmatrix} \qquad (k_{18} = 3)$$

$$= \big(4+\eta_2+3\eta_3-\eta_4\big)I + \big(-3-\eta_1-\eta_2-2\eta_3\big)B$$

$$\rho(T_{19}) = \begin{pmatrix} \frac{1}{2}\eta_4 & 5+\eta_1+2\eta_2+4\eta_3 \\ -\frac{1}{4}\eta_1+\frac{1}{4}\eta_2 & -\frac{1}{2}\eta_4 \end{pmatrix} \qquad (k_{19}=3)$$

$$= \big(-10-2\eta_1-3\eta_2-7\eta_3+\eta_4\big)A + \big(5+\eta_1+2\eta_2+4\eta_3\big)AB$$

$$\rho(T_{20}) = \begin{pmatrix} \frac{1}{2}\eta_4 & -5-\eta_1-2\eta_2-4\eta_3 \\ -\frac{1}{4}\eta_1+\frac{1}{4}\eta_2 & \frac{1}{2}\eta_4 \end{pmatrix} \qquad (k_{20}=3)$$

$$= \big(10+2\eta_1+3\eta_2+7\eta_3\big)I + \big(-5-\eta_1-2\eta_2-4\eta_3\big)B$$

$$\rho(T_{21}) = \begin{pmatrix} \frac{1}{2}\eta_4 & 5+\eta_1+2\eta_2+4\eta_3 \\ \frac{1}{4}\eta_1-\frac{1}{4}\eta_2 & \frac{1}{2}\eta_4 \end{pmatrix} \qquad (k_{21}=3)$$

$$= \big(-10-2\eta_1-3\eta_2-7\eta_3+\eta_4\big)I + \big(5+\eta_1+2\eta_2+4\eta_3\big)B$$

$$\rho(T_{22}) = \begin{pmatrix} \frac{1}{2}\eta_4 & -5-\eta_1-2\eta_2-4\eta_3 \\ \frac{1}{4}\eta_1-\frac{1}{4}\eta_2 & -\frac{1}{2}\eta_4 \end{pmatrix} \qquad (k_{22}=3)$$

$$= \big(10+2\eta_1+3\eta_2+7\eta_3\big)A + \big(-5-\eta_1-2\eta_2-4\eta_3\big)AB \qquad .$$

Block 2: $M_2(\mathbb{C})$ on χ_8 (degree 6)

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} \frac{1}{2}\eta_1 & 1 \\ -\frac{3}{2}+\frac{1}{2}\eta_1+\frac{1}{4}\eta_2-\frac{1}{2}\eta_4 & \frac{1}{2}\eta_1 \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$A^2 - I = 0,$$
$$B^2 - \eta_1\, B + \big(3+\eta_4\big)I = 0,$$
$$BA = \eta_1\, A - AB,$$
$$ABA = \eta_1 I - B,$$
$$ABB = \big(-3-\eta_4\big)A + \eta_1\, AB.$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \qquad (k_1=1)$$

$$\rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A \qquad (k_2=1)$$

$$\rho(T_3) = \begin{pmatrix} \frac{1}{2}\eta_1 & 1 \\ -\frac{3}{2}+\frac{1}{2}\eta_1+\frac{1}{4}\eta_2-\frac{1}{2}\eta_4 & \frac{1}{2}\eta_1 \end{pmatrix} = B \qquad (k_3=3)$$

$$\rho(T_4) = \begin{pmatrix} \frac{1}{2}\eta_1 & -1 \\ -\frac{3}{2}+\frac{1}{2}\eta_1+\frac{1}{4}\eta_2-\frac{1}{2}\eta_4 & -\frac{1}{2}\eta_1 \end{pmatrix} = \eta_1\, A - AB \qquad (k_4=3)$$

$$\rho(T_5) = \begin{pmatrix} \frac{1}{2}\eta_2 & 2+\eta_1 \\ -\frac{1}{4}\eta_1+\frac{1}{2}\eta_4-\frac{1}{4}\eta_8 & \frac{1}{2}\eta_2 \end{pmatrix} \qquad (k_5=3)$$

$$= \big(-3-2\eta_1-\eta_4\big)I + \big(2+\eta_1\big)B$$

$$\rho(T_6) = \begin{pmatrix} \frac{1}{2}\eta_2 & -2-\eta_1 \\ -\frac{1}{4}\eta_1+\frac{1}{2}\eta_4-\frac{1}{4}\eta_8 & -\frac{1}{2}\eta_2 \end{pmatrix} \qquad (k_6=3)$$

$$= \big(3+2\eta_1+\eta_2+\eta_4\big)A + \big(-2-\eta_1\big)AB$$

$$\rho(T_7) = \begin{pmatrix} \frac{1}{2}\eta_1 & 1 \\ \frac{3}{2} - \frac{1}{2}\eta_1 - \frac{1}{4}\eta_2 + \frac{1}{2}\eta_4 & -\frac{1}{2}\eta_1 \end{pmatrix} = AB \qquad (k_7 = 3)$$

$$\rho(T_8) = \begin{pmatrix} \frac{1}{2}\eta_1 & -1 \\ \frac{3}{2} - \frac{1}{2}\eta_1 - \frac{1}{4}\eta_2 + \frac{1}{2}\eta_4 & \frac{1}{2}\eta_1 \end{pmatrix} = \eta_1 I - B \qquad (k_8 = 3)$$

$$\rho(T_9) = \begin{pmatrix} \frac{1}{2}\eta_3 & 3 + 2\eta_1 + \eta_2 + \eta_4 \\ -\frac{1}{2}\eta_2 + \frac{1}{4}\eta_3 + \frac{1}{4}\eta_4 & \frac{1}{2}\eta_3 \end{pmatrix} \qquad (k_9 = 3)$$
$$= \big(-5 - 4\eta_1 - \eta_2 - 2\eta_4\big)I + \big(3 + 2\eta_1 + \eta_2 + \eta_4\big)B$$

$$\rho(T_{10}) = \begin{pmatrix} \frac{1}{2}\eta_3 & -3 - 2\eta_1 - \eta_2 - \eta_4 \\ -\frac{1}{2}\eta_2 + \frac{1}{4}\eta_3 + \frac{1}{4}\eta_4 & -\frac{1}{2}\eta_3 \end{pmatrix} \qquad (k_{10} = 3)$$
$$= \big(4 + 3\eta_1 + \eta_4 - \eta_8\big)A + \big(-3 - 2\eta_1 - \eta_2 - \eta_4\big)AB$$

$$\rho(T_{11}) = \begin{pmatrix} \frac{1}{2}\eta_2 & 2 + \eta_1 \\ \frac{1}{4}\eta_1 - \frac{1}{2}\eta_4 + \frac{1}{4}\eta_8 & -\frac{1}{2}\eta_2 \end{pmatrix} \qquad (k_{11} = 3)$$
$$= \big(-3 - 2\eta_1 - \eta_4\big)A + \big(2 + \eta_1\big)AB$$

$$\rho(T_{12}) = \begin{pmatrix} \frac{1}{2}\eta_2 & -2 - \eta_1 \\ \frac{1}{4}\eta_1 - \frac{1}{2}\eta_4 + \frac{1}{4}\eta_8 & \frac{1}{2}\eta_2 \end{pmatrix} \qquad (k_{12} = 3)$$
$$= \big(3 + 2\eta_1 + \eta_2 + \eta_4\big)I + \big(-2 - \eta_1\big)B$$

$$\rho(T_{13}) = \begin{pmatrix} \frac{1}{2}\eta_4 & 2 + \eta_1 - \eta_8 \\ \frac{1}{4}\eta_3 - \frac{1}{4}\eta_8 & -\frac{1}{2}\eta_4 \end{pmatrix} \qquad (k_{13} = 3)$$
$$= \big(-4 - 3\eta_1 - \eta_2 - \eta_4\big)A + \big(2 + \eta_1 - \eta_8\big)AB$$

$$\rho(T_{14}) = \begin{pmatrix} \frac{1}{2}\eta_4 & -2 - \eta_1 + \eta_8 \\ \frac{1}{4}\eta_3 - \frac{1}{4}\eta_8 & \frac{1}{2}\eta_4 \end{pmatrix} \qquad (k_{14} = 3)$$
$$= \big(4 + 3\eta_1 + \eta_2 + 2\eta_4\big)I + \big(-2 - \eta_1 + \eta_8\big)B$$

$$\rho(T_{15}) = \begin{pmatrix} \frac{1}{2}\eta_4 & 2 + \eta_1 - \eta_8 \\ -\frac{1}{4}\eta_3 + \frac{1}{4}\eta_8 & \frac{1}{2}\eta_4 \end{pmatrix} \qquad (k_{15} = 3)$$
$$= \big(-4 - 3\eta_1 - \eta_2 - \eta_4\big)I + \big(2 + \eta_1 - \eta_8\big)B$$

$$\rho(T_{16}) = \begin{pmatrix} \frac{1}{2}\eta_4 & -2 - \eta_1 + \eta_8 \\ -\frac{1}{4}\eta_3 + \frac{1}{4}\eta_8 & -\frac{1}{2}\eta_4 \end{pmatrix} \qquad (k_{16} = 3)$$
$$= \big(4 + 3\eta_1 + \eta_2 + 2\eta_4\big)A + \big(-2 - \eta_1 + \eta_8\big)AB$$

$$\rho(T_{17}) = \begin{pmatrix} \frac{1}{2}\eta_3 & 3 + 2\eta_1 + \eta_2 + \eta_4 \\ \frac{1}{2}\eta_2 - \frac{1}{4}\eta_3 - \frac{1}{4}\eta_4 & -\frac{1}{2}\eta_3 \end{pmatrix} \qquad (k_{17} = 3)$$
$$= \big(-5 - 4\eta_1 - \eta_2 - 2\eta_4\big)A + \big(3 + 2\eta_1 + \eta_2 + \eta_4\big)AB$$

$$\rho(T_{18}) = \begin{pmatrix} \frac{1}{2}\eta_3 & -3 - 2\eta_1 - \eta_2 - \eta_4 \\ \frac{1}{2}\eta_2 - \frac{1}{4}\eta_3 - \frac{1}{4}\eta_4 & \frac{1}{2}\eta_3 \end{pmatrix} \qquad (k_{18} = 3)$$
$$= \big(4 + 3\eta_1 + \eta_4 - \eta_8\big)I + \big(-3 - 2\eta_1 - \eta_2 - \eta_4\big)B$$

$$\rho(T_{19}) = \begin{pmatrix} \frac{1}{2}\eta_8 & 5 + 4\eta_1 + \eta_2 + 2\eta_4 \\ -\frac{1}{4}\eta_2 + \frac{1}{4}\eta_4 & -\frac{1}{2}\eta_8 \end{pmatrix} \qquad (k_{19} = 3)$$
$$= \big(-11 - 8\eta_1 - 3\eta_2 - \eta_3 - 4\eta_4\big)A + \big(5 + 4\eta_1 + \eta_2 + 2\eta_4\big)AB$$

$$\rho(T_{20}) = \begin{pmatrix} \frac{1}{2}\eta_8 & -5 - 4\eta_1 - \eta_2 - 2\eta_4 \\ -\frac{1}{4}\eta_2 + \frac{1}{4}\eta_4 & \frac{1}{2}\eta_8 \end{pmatrix} \qquad (k_{20} = 3)$$
$$= \big(10 + 7\eta_1 + 2\eta_2 + 3\eta_4\big)I + \big(-5 - 4\eta_1 - \eta_2 - 2\eta_4\big)B$$

$$\rho(T_{21}) = \begin{pmatrix} \frac{1}{2}\eta_8 & 5+4\eta_1+\eta_2+2\eta_4 \\ \frac{1}{4}\eta_2-\frac{1}{4}\eta_4 & \frac{1}{2}\eta_8 \end{pmatrix} \qquad (k_{21}=3)$$
$$= \big(-11-8\eta_1-3\eta_2-\eta_3-4\eta_4\big)I + \big(5+4\eta_1+\eta_2+2\eta_4\big)B$$
$$\rho(T_{22}) = \begin{pmatrix} \frac{1}{2}\eta_8 & -5-4\eta_1-\eta_2-2\eta_4 \\ \frac{1}{4}\eta_2-\frac{1}{4}\eta_4 & -\frac{1}{2}\eta_8 \end{pmatrix} \qquad (k_{22}=3)$$
$$= \big(10+7\eta_1+2\eta_2+3\eta_4\big)A + \big(-5-4\eta_1-\eta_2-2\eta_4\big)AB \qquad .$$

Block 3: $M_2(\mathbb{C})$ on χ_9 (degree 6)

Generators.

$$A=\rho(T_2)=\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}, \qquad B=\rho(T_3)=\begin{pmatrix}\frac{1}{2}\eta_8 & 1\\ -\frac{3}{2}-\frac{1}{2}\eta_1+\frac{1}{4}\eta_3+\frac{1}{2}\eta_8 & \frac{1}{2}\eta_8\end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$A^2 - I = 0,$$
$$B^2 - \eta_8\, B + \big(3+\eta_1\big)I = 0,$$
$$BA = \eta_8\, A - AB,$$
$$ABA = \eta_8 I - B,$$
$$ABB = \big(-3-\eta_1\big)A + \eta_8\, AB.$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} = I \qquad (k_1=1)$$
$$\rho(T_2) = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} = A \qquad (k_2=1)$$
$$\rho(T_3) = \begin{pmatrix}\frac{1}{2}\eta_8 & 1\\ -\frac{3}{2}-\frac{1}{2}\eta_1+\frac{1}{4}\eta_3+\frac{1}{2}\eta_8 & \frac{1}{2}\eta_8\end{pmatrix} = B \qquad (k_3=3)$$
$$\rho(T_4) = \begin{pmatrix}\frac{1}{2}\eta_8 & -1\\ -\frac{3}{2}-\frac{1}{2}\eta_1+\frac{1}{4}\eta_3+\frac{1}{2}\eta_8 & -\frac{1}{2}\eta_8\end{pmatrix} = \eta_8\, A - AB \qquad (k_4=3)$$
$$\rho(T_5) = \begin{pmatrix}\frac{1}{2}\eta_3 & 2+\eta_8\\ \frac{1}{2}\eta_1-\frac{1}{4}\eta_2-\frac{1}{4}\eta_8 & \frac{1}{2}\eta_3\end{pmatrix} \qquad (k_5=3)$$
$$= \big(-3-\eta_1-2\eta_8\big)I + \big(2+\eta_8\big)B$$
$$\rho(T_6) = \begin{pmatrix}\frac{1}{2}\eta_3 & -2-\eta_8\\ \frac{1}{2}\eta_1-\frac{1}{4}\eta_2-\frac{1}{4}\eta_8 & -\frac{1}{2}\eta_3\end{pmatrix} \qquad (k_6=3)$$
$$= \big(3+\eta_1+\eta_3+2\eta_8\big)A + \big(-2-\eta_8\big)AB$$
$$\rho(T_7) = \begin{pmatrix}\frac{1}{2}\eta_8 & 1\\ \frac{3}{2}+\frac{1}{2}\eta_1-\frac{1}{4}\eta_3-\frac{1}{2}\eta_8 & -\frac{1}{2}\eta_8\end{pmatrix} = AB \qquad (k_7=3)$$
$$\rho(T_8) = \begin{pmatrix}\frac{1}{2}\eta_8 & -1\\ \frac{3}{2}+\frac{1}{2}\eta_1-\frac{1}{4}\eta_3-\frac{1}{2}\eta_8 & \frac{1}{2}\eta_8\end{pmatrix} = \eta_8 I - B \qquad (k_8=3)$$
$$\rho(T_9) = \begin{pmatrix}\frac{1}{2}\eta_4 & 3+\eta_1+\eta_3+2\eta_8\\ \frac{1}{4}\eta_1-\frac{1}{2}\eta_3+\frac{1}{4}\eta_4 & \frac{1}{2}\eta_4\end{pmatrix} \qquad (k_9=3)$$
$$= \big(-5-2\eta_1-\eta_3-4\eta_8\big)I + \big(3+\eta_1+\eta_3+2\eta_8\big)B$$

$$
\begin{aligned}
\rho(T_{10}) &= \begin{pmatrix} \frac{1}{2}\eta_4 & -3-\eta_1-\eta_3-2\eta_8 \\ \frac{1}{4}\eta_1-\frac{1}{2}\eta_3+\frac{1}{4}\eta_4 & -\frac{1}{2}\eta_4 \end{pmatrix} && (k_{10}=3)\\
&= \big(4+\eta_1-\eta_2+3\eta_8\big)A + \big(-3-\eta_1-\eta_3-2\eta_8\big)AB \\
\rho(T_{11}) &= \begin{pmatrix} \frac{1}{2}\eta_3 & 2+\eta_8 \\ -\frac{1}{2}\eta_1+\frac{1}{4}\eta_2+\frac{1}{4}\eta_8 & -\frac{1}{2}\eta_3 \end{pmatrix} && (k_{11}=3)\\
&= \big(-3-\eta_1-2\eta_8\big)A + \big(2+\eta_8\big)AB \\
\rho(T_{12}) &= \begin{pmatrix} \frac{1}{2}\eta_3 & -2-\eta_8 \\ -\frac{1}{2}\eta_1+\frac{1}{4}\eta_2+\frac{1}{4}\eta_8 & \frac{1}{2}\eta_3 \end{pmatrix} && (k_{12}=3)\\
&= \big(3+\eta_1+\eta_3+2\eta_8\big)I + \big(-2-\eta_8\big)B \\
\rho(T_{13}) &= \begin{pmatrix} \frac{1}{2}\eta_1 & 2-\eta_2+\eta_8 \\ -\frac{1}{4}\eta_2+\frac{1}{4}\eta_4 & -\frac{1}{2}\eta_1 \end{pmatrix} && (k_{13}=3)\\
&= \big(-4-\eta_1-\eta_3-3\eta_8\big)A + \big(2-\eta_2+\eta_8\big)AB \\
\rho(T_{14}) &= \begin{pmatrix} \frac{1}{2}\eta_1 & -2+\eta_2-\eta_8 \\ -\frac{1}{4}\eta_2+\frac{1}{4}\eta_4 & \frac{1}{2}\eta_1 \end{pmatrix} && (k_{14}=3)\\
&= \big(4+2\eta_1+\eta_3+3\eta_8\big)I + \big(-2+\eta_2-\eta_8\big)B \\
\rho(T_{15}) &= \begin{pmatrix} \frac{1}{2}\eta_1 & 2-\eta_2+\eta_8 \\ \frac{1}{4}\eta_2-\frac{1}{4}\eta_4 & \frac{1}{2}\eta_1 \end{pmatrix} && (k_{15}=3)\\
&= \big(-4-\eta_1-\eta_3-3\eta_8\big)I + \big(2-\eta_2+\eta_8\big)B \\
\rho(T_{16}) &= \begin{pmatrix} \frac{1}{2}\eta_1 & -2+\eta_2-\eta_8 \\ \frac{1}{4}\eta_2-\frac{1}{4}\eta_4 & -\frac{1}{2}\eta_1 \end{pmatrix} && (k_{16}=3)\\
&= \big(4+2\eta_1+\eta_3+3\eta_8\big)A + \big(-2+\eta_2-\eta_8\big)AB \\
\rho(T_{17}) &= \begin{pmatrix} \frac{1}{2}\eta_4 & 3+\eta_1+\eta_3+2\eta_8 \\ -\frac{1}{4}\eta_1+\frac{1}{2}\eta_3-\frac{1}{4}\eta_4 & -\frac{1}{2}\eta_4 \end{pmatrix} && (k_{17}=3)\\
&= \big(-5-2\eta_1-\eta_3-4\eta_8\big)A + \big(3+\eta_1+\eta_3+2\eta_8\big)AB \\
\rho(T_{18}) &= \begin{pmatrix} \frac{1}{2}\eta_4 & -3-\eta_1-\eta_3-2\eta_8 \\ -\frac{1}{4}\eta_1+\frac{1}{2}\eta_3-\frac{1}{4}\eta_4 & \frac{1}{2}\eta_4 \end{pmatrix} && (k_{18}=3)\\
&= \big(4+\eta_1-\eta_2+3\eta_8\big)I + \big(-3-\eta_1-\eta_3-2\eta_8\big)B \\
\rho(T_{19}) &= \begin{pmatrix} \frac{1}{2}\eta_2 & 5+2\eta_1+\eta_3+4\eta_8 \\ \frac{1}{4}\eta_1-\frac{1}{4}\eta_3 & -\frac{1}{2}\eta_2 \end{pmatrix} && (k_{19}=3)\\
&= \big(-3+4\eta_1+8\eta_2+5\eta_3+7\eta_4\big)A + \big(5+2\eta_1+\eta_3+4\eta_8\big)AB \\
\rho(T_{20}) &= \begin{pmatrix} \frac{1}{2}\eta_2 & -5-2\eta_1-\eta_3-4\eta_8 \\ \frac{1}{4}\eta_1-\frac{1}{4}\eta_3 & \frac{1}{2}\eta_2 \end{pmatrix} && (k_{20}=3)\\
&= \big(10+3\eta_1+2\eta_3+7\eta_8\big)I + \big(-5-2\eta_1-\eta_3-4\eta_8\big)B \\
\rho(T_{21}) &= \begin{pmatrix} \frac{1}{2}\eta_2 & 5+2\eta_1+\eta_3+4\eta_8 \\ -\frac{1}{4}\eta_1+\frac{1}{4}\eta_3 & \frac{1}{2}\eta_2 \end{pmatrix} && (k_{21}=3)\\
&= \big(-3+4\eta_1+8\eta_2+5\eta_3+7\eta_4\big)I + \big(5+2\eta_1+\eta_3+4\eta_8\big)B \\
\rho(T_{22}) &= \begin{pmatrix} \frac{1}{2}\eta_2 & -5-2\eta_1-\eta_3-4\eta_8 \\ -\frac{1}{4}\eta_1+\frac{1}{4}\eta_3 & -\frac{1}{2}\eta_2 \end{pmatrix} && (k_{22}=3)\\
&= \big(10+3\eta_1+2\eta_3+7\eta_8\big)A + \big(-5-2\eta_1-\eta_3-4\eta_8\big)AB && .
\end{aligned}
$$

Block 4: $M_2(\mathbb{C})$ on χ_{10} (degree 6)

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} \frac{1}{2}\eta_2 & 1 \\ -\frac{3}{2} + \frac{1}{2}\eta_2 + \frac{1}{4}\eta_4 - \frac{1}{2}\eta_8 & \frac{1}{2}\eta_2 \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$\begin{gathered} A^2 - I = 0, \\ B^2 - \eta_2\, B + \big(3 + \eta_8\big) I = 0, \\ BA = \eta_2\, A - AB, \\ ABA = \eta_2 I - B, \\ ABB = \big(-3 - \eta_8\big) A + \eta_2\, AB. \end{gathered}$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$

$$\rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A \qquad (k_2 = 1)$$

$$\rho(T_3) = \begin{pmatrix} \frac{1}{2}\eta_2 & 1 \\ -\frac{3}{2} + \frac{1}{2}\eta_2 + \frac{1}{4}\eta_4 - \frac{1}{2}\eta_8 & \frac{1}{2}\eta_2 \end{pmatrix} = B \qquad (k_3 = 3)$$

$$\rho(T_4) = \begin{pmatrix} \frac{1}{2}\eta_2 & -1 \\ -\frac{3}{2} + \frac{1}{2}\eta_2 + \frac{1}{4}\eta_4 - \frac{1}{2}\eta_8 & -\frac{1}{2}\eta_2 \end{pmatrix} = \eta_2\, A - AB \qquad (k_4 = 3)$$

$$\begin{aligned} \rho(T_5) &= \begin{pmatrix} \frac{1}{2}\eta_4 & 2 + \eta_2 \\ -\frac{1}{4}\eta_2 - \frac{1}{4}\eta_3 + \frac{1}{2}\eta_8 & \frac{1}{2}\eta_4 \end{pmatrix} \qquad (k_5 = 3) \\ &= \big(-3 - 2\eta_2 - \eta_8\big) I + \big(2 + \eta_2\big) B \end{aligned}$$

$$\begin{aligned} \rho(T_6) &= \begin{pmatrix} \frac{1}{2}\eta_4 & -2 - \eta_2 \\ -\frac{1}{4}\eta_2 - \frac{1}{4}\eta_3 + \frac{1}{2}\eta_8 & -\frac{1}{2}\eta_4 \end{pmatrix} \qquad (k_6 = 3) \\ &= \big(2 - \eta_1 + \eta_2 - \eta_3\big) A + \big(-2 - \eta_2\big) AB \end{aligned}$$

$$\rho(T_7) = \begin{pmatrix} \frac{1}{2}\eta_2 & 1 \\ \frac{3}{2} - \frac{1}{2}\eta_2 - \frac{1}{4}\eta_4 + \frac{1}{2}\eta_8 & -\frac{1}{2}\eta_2 \end{pmatrix} = AB \qquad (k_7 = 3)$$

$$\rho(T_8) = \begin{pmatrix} \frac{1}{2}\eta_2 & -1 \\ \frac{3}{2} - \frac{1}{2}\eta_2 - \frac{1}{4}\eta_4 + \frac{1}{2}\eta_8 & \frac{1}{2}\eta_2 \end{pmatrix} = \eta_2 I - B \qquad (k_8 = 3)$$

$$\begin{aligned} \rho(T_9) &= \begin{pmatrix} \frac{1}{2}\eta_1 & 2 - \eta_1 + \eta_2 - \eta_3 \\ \frac{1}{4}\eta_1 - \frac{1}{2}\eta_4 + \frac{1}{4}\eta_8 & \frac{1}{2}\eta_1 \end{pmatrix} \qquad (k_9 = 3) \\ &= \big(-5 - 4\eta_2 - \eta_4 - 2\eta_8\big) I + \big(2 - \eta_1 + \eta_2 - \eta_3\big) B \end{aligned}$$

$$\begin{aligned} \rho(T_{10}) &= \begin{pmatrix} \frac{1}{2}\eta_1 & -2 + \eta_1 - \eta_2 + \eta_3 \\ \frac{1}{4}\eta_1 - \frac{1}{2}\eta_4 + \frac{1}{4}\eta_8 & -\frac{1}{2}\eta_1 \end{pmatrix} \qquad (k_{10} = 3) \\ &= \big(4 + 3\eta_2 - \eta_3 + \eta_8\big) A + \big(-2 + \eta_1 - \eta_2 + \eta_3\big) AB \end{aligned}$$

$$\begin{aligned} \rho(T_{11}) &= \begin{pmatrix} \frac{1}{2}\eta_4 & 2 + \eta_2 \\ \frac{1}{4}\eta_2 + \frac{1}{4}\eta_3 - \frac{1}{2}\eta_8 & -\frac{1}{2}\eta_4 \end{pmatrix} \qquad (k_{11} = 3) \\ &= \big(-3 - 2\eta_2 - \eta_8\big) A + \big(2 + \eta_2\big) AB \end{aligned}$$

$$\rho(T_{12}) = \begin{pmatrix} \frac{1}{2}\eta_4 & -2 - \eta_2 \\ \frac{1}{4}\eta_2 + \frac{1}{4}\eta_3 - \frac{1}{2}\eta_8 & \frac{1}{2}\eta_4 \end{pmatrix} \qquad (k_{12} = 3)$$

$$
\begin{aligned}
&= \big(2-\eta_1+\eta_2-\eta_3\big)I + \big(-2-\eta_2\big)B \\
\rho(T_{13}) &= \begin{pmatrix} \frac{1}{2}\eta_8 & 2+\eta_2-\eta_3 \\ \frac{1}{4}\eta_1-\frac{1}{4}\eta_3 & -\frac{1}{2}\eta_8 \end{pmatrix} && (k_{13}=3) \\
&= \big(-3+\eta_1-2\eta_2+\eta_3\big)A + \big(2+\eta_2-\eta_3\big)AB \\
\rho(T_{14}) &= \begin{pmatrix} \frac{1}{2}\eta_8 & -2-\eta_2+\eta_3 \\ \frac{1}{4}\eta_1-\frac{1}{4}\eta_3 & \frac{1}{2}\eta_8 \end{pmatrix} && (k_{14}=3) \\
&= \big(4+3\eta_2+\eta_4+2\eta_8\big)I + \big(-2-\eta_2+\eta_3\big)B \\
\rho(T_{15}) &= \begin{pmatrix} \frac{1}{2}\eta_8 & 2+\eta_2-\eta_3 \\ -\frac{1}{4}\eta_1+\frac{1}{4}\eta_3 & \frac{1}{2}\eta_8 \end{pmatrix} && (k_{15}=3) \\
&= \big(-3+\eta_1-2\eta_2+\eta_3\big)I + \big(2+\eta_2-\eta_3\big)B \\
\rho(T_{16}) &= \begin{pmatrix} \frac{1}{2}\eta_8 & -2-\eta_2+\eta_3 \\ -\frac{1}{4}\eta_1+\frac{1}{4}\eta_3 & -\frac{1}{2}\eta_8 \end{pmatrix} && (k_{16}=3) \\
&= \big(4+3\eta_2+\eta_4+2\eta_8\big)A + \big(-2-\eta_2+\eta_3\big)AB \\
\rho(T_{17}) &= \begin{pmatrix} \frac{1}{2}\eta_1 & 2-\eta_1+\eta_2-\eta_3 \\ -\frac{1}{4}\eta_1+\frac{1}{2}\eta_4-\frac{1}{4}\eta_8 & -\frac{1}{2}\eta_1 \end{pmatrix} && (k_{17}=3) \\
&= \big(-5-4\eta_2-\eta_4-2\eta_8\big)A + \big(2-\eta_1+\eta_2-\eta_3\big)AB \\
\rho(T_{18}) &= \begin{pmatrix} \frac{1}{2}\eta_1 & -2+\eta_1-\eta_2+\eta_3 \\ -\frac{1}{4}\eta_1+\frac{1}{2}\eta_4-\frac{1}{4}\eta_8 & \frac{1}{2}\eta_1 \end{pmatrix} && (k_{18}=3) \\
&= \big(4+3\eta_2-\eta_3+\eta_8\big)I + \big(-2+\eta_1-\eta_2+\eta_3\big)B \\
\rho(T_{19}) &= \begin{pmatrix} \frac{1}{2}\eta_3 & 5+4\eta_2+\eta_4+2\eta_8 \\ -\frac{1}{4}\eta_4+\frac{1}{4}\eta_8 & -\frac{1}{2}\eta_3 \end{pmatrix} && (k_{19}=3) \\
&= \big(-7+3\eta_1-4\eta_2+4\eta_3+\eta_4\big)A + \big(5+4\eta_2+\eta_4+2\eta_8\big)AB \\
\rho(T_{20}) &= \begin{pmatrix} \frac{1}{2}\eta_3 & -5-4\eta_2-\eta_4-2\eta_8 \\ -\frac{1}{4}\eta_4+\frac{1}{4}\eta_8 & \frac{1}{2}\eta_3 \end{pmatrix} && (k_{20}=3) \\
&= \big(10+7\eta_2+2\eta_4+3\eta_8\big)I + \big(-5-4\eta_2-\eta_4-2\eta_8\big)B \\
\rho(T_{21}) &= \begin{pmatrix} \frac{1}{2}\eta_3 & 5+4\eta_2+\eta_4+2\eta_8 \\ \frac{1}{4}\eta_4-\frac{1}{4}\eta_8 & \frac{1}{2}\eta_3 \end{pmatrix} && (k_{21}=3) \\
&= \big(-7+3\eta_1-4\eta_2+4\eta_3+\eta_4\big)I + \big(5+4\eta_2+\eta_4+2\eta_8\big)B \\
\rho(T_{22}) &= \begin{pmatrix} \frac{1}{2}\eta_3 & -5-4\eta_2-\eta_4-2\eta_8 \\ \frac{1}{4}\eta_4-\frac{1}{4}\eta_8 & -\frac{1}{2}\eta_3 \end{pmatrix} && (k_{22}=3) \\
&= \big(10+7\eta_2+2\eta_4+3\eta_8\big)A + \big(-5-4\eta_2-\eta_4-2\eta_8\big)AB && .
\end{aligned}
$$

Block 5: $M_2(\mathbb{C})$ on χ_{11} (degree 6)

Generators.

$$
A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} \frac{1}{2}\eta_4 & 1 \\ -\frac{3}{2}-\frac{1}{2}\eta_3+\frac{1}{2}\eta_4+\frac{1}{4}\eta_8 & \frac{1}{2}\eta_4 \end{pmatrix}.
$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$
\begin{gathered}
A^2 - I = 0, \\
B^2 - \eta_4\, B + \big(3+\eta_3\big)I = 0, \\
BA = \eta_4\, A - AB,
\end{gathered}
$$

$$ABA = \eta_4 I - B,$$
$$ABB = \big(-3-\eta_3\big)A + \eta_4\, AB.$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$

$$\rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A \qquad (k_2 = 1)$$

$$\rho(T_3) = \begin{pmatrix} \frac{1}{2}\eta_4 & 1 \\ -\frac{3}{2} - \frac{1}{2}\eta_3 + \frac{1}{2}\eta_4 + \frac{1}{4}\eta_8 & \frac{1}{2}\eta_4 \end{pmatrix} = B \qquad (k_3 = 3)$$

$$\rho(T_4) = \begin{pmatrix} \frac{1}{2}\eta_4 & -1 \\ -\frac{3}{2} - \frac{1}{2}\eta_3 + \frac{1}{2}\eta_4 + \frac{1}{4}\eta_8 & -\frac{1}{2}\eta_4 \end{pmatrix} = \eta_4\, A - AB \qquad (k_4 = 3)$$

$$\rho(T_5) = \begin{pmatrix} \frac{1}{2}\eta_8 & 2+\eta_4 \\ -\frac{1}{4}\eta_1 + \frac{1}{2}\eta_3 - \frac{1}{4}\eta_4 & \frac{1}{2}\eta_8 \end{pmatrix} \qquad (k_5 = 3)$$
$$= \big(-3-\eta_3-2\eta_4\big)I + \big(2+\eta_4\big)B$$

$$\rho(T_6) = \begin{pmatrix} \frac{1}{2}\eta_8 & -2-\eta_4 \\ -\frac{1}{4}\eta_1 + \frac{1}{2}\eta_3 - \frac{1}{4}\eta_4 & -\frac{1}{2}\eta_8 \end{pmatrix} \qquad (k_6 = 3)$$
$$= \big(2-\eta_1-\eta_2+\eta_4\big)A + \big(-2-\eta_4\big)AB$$

$$\rho(T_7) = \begin{pmatrix} \frac{1}{2}\eta_4 & 1 \\ \frac{3}{2} + \frac{1}{2}\eta_3 - \frac{1}{2}\eta_4 - \frac{1}{4}\eta_8 & -\frac{1}{2}\eta_4 \end{pmatrix} = AB \qquad (k_7 = 3)$$

$$\rho(T_8) = \begin{pmatrix} \frac{1}{2}\eta_4 & -1 \\ \frac{3}{2} + \frac{1}{2}\eta_3 - \frac{1}{2}\eta_4 - \frac{1}{4}\eta_8 & \frac{1}{2}\eta_4 \end{pmatrix} = \eta_4 I - B \qquad (k_8 = 3)$$

$$\rho(T_9) = \begin{pmatrix} \frac{1}{2}\eta_2 & 2-\eta_1-\eta_2+\eta_4 \\ \frac{1}{4}\eta_2 + \frac{1}{4}\eta_3 - \frac{1}{2}\eta_8 & \frac{1}{2}\eta_2 \end{pmatrix} \qquad (k_9 = 3)$$
$$= \big(-5-2\eta_3-4\eta_4-\eta_8\big)I + \big(2-\eta_1-\eta_2+\eta_4\big)B$$

$$\rho(T_{10}) = \begin{pmatrix} \frac{1}{2}\eta_2 & -2+\eta_1+\eta_2-\eta_4 \\ \frac{1}{4}\eta_2 + \frac{1}{4}\eta_3 - \frac{1}{2}\eta_8 & -\frac{1}{2}\eta_2 \end{pmatrix} \qquad (k_{10} = 3)$$
$$= \big(4-\eta_1+\eta_3+3\eta_4\big)A + \big(-2+\eta_1+\eta_2-\eta_4\big)AB$$

$$\rho(T_{11}) = \begin{pmatrix} \frac{1}{2}\eta_8 & 2+\eta_4 \\ \frac{1}{4}\eta_1 - \frac{1}{2}\eta_3 + \frac{1}{4}\eta_4 & -\frac{1}{2}\eta_8 \end{pmatrix} \qquad (k_{11} = 3)$$
$$= \big(-3-\eta_3-2\eta_4\big)A + \big(2+\eta_4\big)AB$$

$$\rho(T_{12}) = \begin{pmatrix} \frac{1}{2}\eta_8 & -2-\eta_4 \\ \frac{1}{4}\eta_1 - \frac{1}{2}\eta_3 + \frac{1}{4}\eta_4 & \frac{1}{2}\eta_8 \end{pmatrix} \qquad (k_{12} = 3)$$
$$= \big(2-\eta_1-\eta_2+\eta_4\big)I + \big(-2-\eta_4\big)B$$

$$\rho(T_{13}) = \begin{pmatrix} \frac{1}{2}\eta_3 & 2-\eta_1+\eta_4 \\ -\frac{1}{4}\eta_1 + \frac{1}{4}\eta_2 & -\frac{1}{2}\eta_3 \end{pmatrix} \qquad (k_{13} = 3)$$
$$= \big(-3+\eta_1+\eta_2-2\eta_4\big)A + \big(2-\eta_1+\eta_4\big)AB$$

$$\rho(T_{14}) = \begin{pmatrix} \frac{1}{2}\eta_3 & -2+\eta_1-\eta_4 \\ -\frac{1}{4}\eta_1 + \frac{1}{4}\eta_2 & \frac{1}{2}\eta_3 \end{pmatrix} \qquad (k_{14} = 3)$$
$$= \big(4+2\eta_3+3\eta_4+\eta_8\big)I + \big(-2+\eta_1-\eta_4\big)B$$

$$\rho(T_{15}) = \begin{pmatrix} \frac{1}{2}\eta_3 & 2-\eta_1+\eta_4 \\ \frac{1}{4}\eta_1 - \frac{1}{4}\eta_2 & \frac{1}{2}\eta_3 \end{pmatrix} \qquad (k_{15} = 3)$$

$$= \big(-3+\eta_1+\eta_2-2\eta_4\big)I + \big(2-\eta_1+\eta_4\big)B$$

$$\rho(T_{16}) = \begin{pmatrix} \frac{1}{2}\eta_3 & -2+\eta_1-\eta_4 \\ \frac{1}{4}\eta_1-\frac{1}{4}\eta_2 & -\frac{1}{2}\eta_3 \end{pmatrix} \qquad (k_{16}=3)$$

$$= \big(4+2\eta_3+3\eta_4+\eta_8\big)A + \big(-2+\eta_1-\eta_4\big)AB$$

$$\rho(T_{17}) = \begin{pmatrix} \frac{1}{2}\eta_2 & 2-\eta_1-\eta_2+\eta_4 \\ -\frac{1}{4}\eta_2-\frac{1}{4}\eta_3+\frac{1}{2}\eta_8 & -\frac{1}{2}\eta_2 \end{pmatrix} \qquad (k_{17}=3)$$

$$= \big(-5-2\eta_3-4\eta_4-\eta_8\big)A + \big(2-\eta_1-\eta_2+\eta_4\big)AB$$

$$\rho(T_{18}) = \begin{pmatrix} \frac{1}{2}\eta_2 & -2+\eta_1+\eta_2-\eta_4 \\ -\frac{1}{4}\eta_2-\frac{1}{4}\eta_3+\frac{1}{2}\eta_8 & \frac{1}{2}\eta_2 \end{pmatrix} \qquad (k_{18}=3)$$

$$= \big(4-\eta_1+\eta_3+3\eta_4\big)I + \big(-2+\eta_1+\eta_2-\eta_4\big)B$$

$$\rho(T_{19}) = \begin{pmatrix} \frac{1}{2}\eta_1 & 5+2\eta_3+4\eta_4+\eta_8 \\ \frac{1}{4}\eta_3-\frac{1}{4}\eta_8 & -\frac{1}{2}\eta_1 \end{pmatrix} \qquad (k_{19}=3)$$

$$= \big(-8+3\eta_1+2\eta_2-\eta_3-5\eta_4\big)A + \big(5+2\eta_3+4\eta_4+\eta_8\big)AB$$

$$\rho(T_{20}) = \begin{pmatrix} \frac{1}{2}\eta_1 & -5-2\eta_3-4\eta_4-\eta_8 \\ \frac{1}{4}\eta_3-\frac{1}{4}\eta_8 & \frac{1}{2}\eta_1 \end{pmatrix} \qquad (k_{20}=3)$$

$$= \big(10+3\eta_3+7\eta_4+2\eta_8\big)I + \big(-5-2\eta_3-4\eta_4-\eta_8\big)B$$

$$\rho(T_{21}) = \begin{pmatrix} \frac{1}{2}\eta_1 & 5+2\eta_3+4\eta_4+\eta_8 \\ -\frac{1}{4}\eta_3+\frac{1}{4}\eta_8 & \frac{1}{2}\eta_1 \end{pmatrix} \qquad (k_{21}=3)$$

$$= \big(-8+3\eta_1+2\eta_2-\eta_3-5\eta_4\big)I + \big(5+2\eta_3+4\eta_4+\eta_8\big)B$$

$$\rho(T_{22}) = \begin{pmatrix} \frac{1}{2}\eta_1 & -5-2\eta_3-4\eta_4-\eta_8 \\ -\frac{1}{4}\eta_3+\frac{1}{4}\eta_8 & -\frac{1}{2}\eta_1 \end{pmatrix} \qquad (k_{22}=3)$$

$$= \big(10+3\eta_3+7\eta_4+2\eta_8\big)A + \big(-5-2\eta_3-4\eta_4-\eta_8\big)AB \qquad .$$

Remark 1 (A uniform law for the Frobenius family)**.** *In all three graphs F26A, CR(19), CR(31) the traceless valency-3 generator satisfies*

$$T^2 = (3+\eta)\,I$$

where 3 *is the valency and* η *is a sextic Gauss period of* $\mathbb{Q}(\zeta_p)$*. For* $p=13$ *this is* $3+\frac{-1\mp\sqrt{13}}{2} = \frac{5\mp\sqrt{13}}{2}$*, recovering the F26A relation. The "why-N" constant of the whole family is* valency plus a Gauss period*.*

5 Klein graph F056A: $\big(\mathrm{PSL}(2,7),\, C_3\big)$

Graph. F056A, the 56-vertex cubic arc-transitive graph of girth 6 on the cosets $\mathrm{PSL}(2,7)/C_3$; $G = \mathrm{PSL}(2,7)$ ($|G| = 168$), $H = C_3$.

Algebra. Rank 20, $\mathcal{H} \cong \mathbb{C}^3 \oplus M_2(\mathbb{C}) \oplus M_3(\mathbb{C}) \oplus M_2(\mathbb{C})$ (blocks on χ_4, χ_5, χ_6 of degrees $6, 7, 8$).

Highlights. Block 1 is generated by the α-diagonal orbital ($A^2+A+2I=0$; $2=\alpha\bar{\alpha}$ is the prime that splits in $\mathbb{Q}(\sqrt{-7})$). Block 2, the integral M_3, carries the irreducible characteristic polynomial x^3+x-6 — the first genuinely cubic eigenvalue in the catalogue, echoed by $\mathrm{PSL}(2,13)$'s x^3+x-26 below. In Block 3 the order-3 matrix $\left(\begin{smallmatrix} 1/2 & 1 \\ -7/4 & -3/2 \end{smallmatrix}\right)$ appears as $\rho(T_{10})$ — see the Remark after the Klein $\{7,3\}$ section.

One-dimensional characters.

	$\chi_1\,(1)$	$\chi_2\,(3)$	$\chi_3\,(3)$
T_1	1	1	1
T_2	1	-1	-1
T_3	3	$\bar{\alpha}$	α
T_4	3	α	$\bar{\alpha}$
T_5	3	$-\bar{\alpha}$	$-\alpha$
T_6	3	$-\alpha$	$-\bar{\alpha}$
T_7	3	$-\alpha$	$-\bar{\alpha}$
T_8	3	1	1
T_9	3	-1	-1
T_{10}	3	2	2
T_{11}	3	-2	-2
T_{12}	3	$\bar{\alpha}$	α
T_{13}	3	$1+2\alpha$	$-1-2\alpha$
T_{14}	3	α	$\bar{\alpha}$
T_{15}	3	1	1
T_{16}	3	2	2
T_{17}	3	$-\bar{\alpha}$	$-\alpha$
T_{18}	3	-1	-1
T_{19}	3	$-1-2\alpha$	$1+2\alpha$
T_{20}	3	-2	-2

Block 1: $M_2(\mathbb{C})$ on χ_4 (degree 6)

Generators.

$$A=\rho(T_3)=\begin{pmatrix}\alpha & 0\\ 0 & \bar{\alpha}\end{pmatrix},\qquad B=\rho(T_8)=\begin{pmatrix}1 & 2\\ 2 & 1\end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$A^2+A+2I=0,$$
$$B^2-2\,B-3I=0,$$
$$BA=I+2\,A-B-AB,$$
$$ABA=-4I-A+2\,B,$$
$$ABB=3\,A+2\,AB.$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1)=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}=I \qquad (k_1=1)$$
$$\rho(T_2)=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}=I \qquad (k_2=1)$$
$$\rho(T_3)=\begin{pmatrix}\alpha & 0\\ 0 & \bar{\alpha}\end{pmatrix}=A \qquad (k_3=3)$$
$$\rho(T_4)=\begin{pmatrix}\bar{\alpha} & 0\\ 0 & \alpha\end{pmatrix}=-I-A \qquad (k_4=3)$$

$$\rho(T_5) = \begin{pmatrix} \alpha & 0 \\ 0 & \bar{\alpha} \end{pmatrix} = A \qquad (k_5 = 3)$$

$$\rho(T_6) = \begin{pmatrix} \bar{\alpha} & 0 \\ 0 & \alpha \end{pmatrix} = -I - A \qquad (k_6 = 3)$$

$$\rho(T_7) = \begin{pmatrix} \bar{\alpha} & 0 \\ 0 & \alpha \end{pmatrix} = -I - A \qquad (k_7 = 3)$$

$$\rho(T_8) = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} = B \qquad (k_8 = 3)$$

$$\rho(T_9) = \begin{pmatrix} 1 & -1+\alpha \\ -2-\alpha & 1 \end{pmatrix} = \tfrac{3}{2} I - \tfrac{1}{2} A - \tfrac{1}{2} B + \tfrac{1}{2} AB \qquad (k_9 = 3)$$

$$\rho(T_{10}) = \begin{pmatrix} 0 & \bar{\alpha} \\ \alpha & 0 \end{pmatrix} = \tfrac{1}{2} I + \tfrac{1}{2} A - \tfrac{1}{2} B - \tfrac{1}{2} AB \qquad (k_{10} = 3)$$

$$\rho(T_{11}) = \begin{pmatrix} 0 & \bar{\alpha} \\ \alpha & 0 \end{pmatrix} = \tfrac{1}{2} I + \tfrac{1}{2} A - \tfrac{1}{2} B - \tfrac{1}{2} AB \qquad (k_{11} = 3)$$

$$\rho(T_{12}) = \begin{pmatrix} \alpha & 0 \\ 0 & \bar{\alpha} \end{pmatrix} = A \qquad (k_{12} = 3)$$

$$\rho(T_{13}) = \begin{pmatrix} -1 & -\bar{\alpha} \\ -\alpha & -1 \end{pmatrix} = -\tfrac{3}{2} I - \tfrac{1}{2} A + \tfrac{1}{2} B + \tfrac{1}{2} AB \qquad (k_{13} = 3)$$

$$\rho(T_{14}) = \begin{pmatrix} \bar{\alpha} & 0 \\ 0 & \alpha \end{pmatrix} = -I - A \qquad (k_{14} = 3)$$

$$\rho(T_{15}) = \begin{pmatrix} 1 & -1+\alpha \\ -2-\alpha & 1 \end{pmatrix} = \tfrac{3}{2} I - \tfrac{1}{2} A - \tfrac{1}{2} B + \tfrac{1}{2} AB \qquad (k_{15} = 3)$$

$$\rho(T_{16}) = \begin{pmatrix} 0 & \bar{\alpha} \\ \alpha & 0 \end{pmatrix} = \tfrac{1}{2} I + \tfrac{1}{2} A - \tfrac{1}{2} B - \tfrac{1}{2} AB \qquad (k_{16} = 3)$$

$$\rho(T_{17}) = \begin{pmatrix} \alpha & 0 \\ 0 & \bar{\alpha} \end{pmatrix} = A \qquad (k_{17} = 3)$$

$$\rho(T_{18}) = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} = B \qquad (k_{18} = 3)$$

$$\rho(T_{19}) = \begin{pmatrix} -1 & -\bar{\alpha} \\ -\alpha & -1 \end{pmatrix} = -\tfrac{3}{2} I - \tfrac{1}{2} A + \tfrac{1}{2} B + \tfrac{1}{2} AB \qquad (k_{19} = 3)$$

$$\rho(T_{20}) = \begin{pmatrix} 0 & \bar{\alpha} \\ \alpha & 0 \end{pmatrix} = \tfrac{1}{2} I + \tfrac{1}{2} A - \tfrac{1}{2} B - \tfrac{1}{2} AB \qquad (k_{20} = 3).$$

Block 2: $M_3(\mathbb{C})$ on χ_5 (degree 7)

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 2 \\ 4 & 0 & 1 \end{pmatrix}.$$

Complete relations (monomial basis I, A, B, AB, BA, BB, ABB, BAB, $BABB$ of $M_3(\mathbb{C})$):

$$A^2 - I = 0,$$
$$B^3 - 2\,B^2 - B - 6I = 0,$$
$$ABA = 2I + 2\,A - B - AB - BA,$$

$$
\begin{gathered}
BAA = B,\\
BBA = I + A + 2\,B + 2\,AB + 2\,BA - BB - ABB - BAB,\\
BBB = 6I + B + 2\,BB,\\
ABBA = 5I + 5\,A - 2\,B - 2\,AB - 2\,BA + BAB,\\
ABBB = 6\,A + AB + 2\,ABB,\\
BABA = -I - A - 2\,AB + ABB,\\
BABBA = -8I - 8\,A + B - 4\,AB + BA + 2\,ABB + 2\,BAB - BABB,\\
BABBB = 6\,BA + BAB + 2\,BABB.
\end{gathered}
$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$

$$\rho(T_2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = A \qquad (k_2 = 1)$$

$$\rho(T_3) = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 2 \\ 4 & 0 & 1 \end{pmatrix} = B \qquad (k_3 = 3)$$

$$\rho(T_4) = \begin{pmatrix} 1 & 0 & 1 \\ 4 & 1 & 0 \\ 2 & 2 & 0 \end{pmatrix} = -\tfrac{1}{2}\,B + \tfrac{1}{2}\,BB \qquad (k_4 = 3)$$

$$\rho(T_5) = \begin{pmatrix} 1 & 1 & 0 \\ -2 & 0 & -2 \\ -4 & 0 & -1 \end{pmatrix} = AB \qquad (k_5 = 3)$$

$$\rho(T_6) = \begin{pmatrix} 1 & 0 & 1 \\ -4 & -1 & 0 \\ -2 & -2 & 0 \end{pmatrix} = -\tfrac{1}{2}\,AB + \tfrac{1}{2}\,ABB \qquad (k_6 = 3)$$

$$\rho(T_7) = \begin{pmatrix} 1 & 0 & -1 \\ 4 & -1 & 0 \\ 2 & -2 & 0 \end{pmatrix} \qquad (k_7 = 3)$$

$$= \tfrac{1}{2}I + \tfrac{1}{2}\,A + B + AB + \tfrac{1}{2}\,BA - \tfrac{1}{2}\,BB - \tfrac{1}{2}\,ABB - \tfrac{1}{2}\,BAB$$

$$\rho(T_8) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & -3 \end{pmatrix} \qquad (k_8 = 3)$$

$$= -\tfrac{7}{4}I - \tfrac{1}{4}\,A - \tfrac{1}{2}\,B - AB - \tfrac{1}{4}\,BA + \tfrac{1}{4}\,BB + \tfrac{1}{4}\,ABB + \tfrac{1}{2}\,BAB - \tfrac{1}{4}\,BABB$$

$$\rho(T_9) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 3 & -2 \\ 0 & 0 & -1 \end{pmatrix} \qquad (k_9 = 3)$$

$$= -I - \tfrac{5}{2}\,A - \tfrac{1}{4}\,AB + \tfrac{1}{2}\,BA + \tfrac{1}{4}\,ABB + \tfrac{1}{4}\,BAB - \tfrac{1}{4}\,BABB$$

$$\rho(T_{10}) = \begin{pmatrix} -1 & -1 & 1 \\ 2 & 0 & 0 \\ -2 & 0 & 0 \end{pmatrix} \qquad (k_{10} = 3)$$

$$= -\tfrac{1}{4}I - \tfrac{1}{4}\,A - B - \tfrac{1}{4}\,AB - \tfrac{1}{4}\,BA + \tfrac{1}{4}\,BB - \tfrac{1}{4}\,BAB$$

$$\rho(T_{11}) = \begin{pmatrix} -1 & 1 & -1 \\ 2 & 0 & 0 \\ -2 & 0 & 0 \end{pmatrix} \qquad (k_{11} = 3)$$

$$= -\tfrac{1}{4}I - \tfrac{1}{4}\,A + \tfrac{1}{2}\,B + \tfrac{5}{4}\,AB - \tfrac{1}{4}\,BA - \tfrac{1}{4}\,BB - \tfrac{1}{2}\,ABB - \tfrac{1}{4}\,BAB$$

$$\rho(T_{12}) = \begin{pmatrix} 1 & -1 & 0 \\ -2 & 0 & 2 \\ -4 & 0 & 1 \end{pmatrix} = 2I + 2\,A - B - AB - BA \qquad (k_{12} = 3)$$

$$\rho(T_{13}) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 2 \\ 0 & -2 & 1 \end{pmatrix} \qquad (k_{13} = 3)$$

$$= -\tfrac{1}{4}I - \tfrac{1}{4}\,A + \tfrac{1}{2}\,B - \tfrac{1}{4}\,AB - \tfrac{1}{4}\,BA - \tfrac{1}{4}\,BB - \tfrac{1}{4}\,BAB$$

$$\rho(T_{14}) = \begin{pmatrix} 1 & 0 & -1 \\ -4 & 1 & 0 \\ -2 & 2 & 0 \end{pmatrix} = \tfrac{3}{2}I + \tfrac{3}{2}\,A - \tfrac{1}{2}\,B - \tfrac{1}{2}\,AB - \tfrac{1}{2}\,BA + \tfrac{1}{2}\,BAB \qquad (k_{14} = 3)$$

$$\rho(T_{15}) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -3 & 2 \\ 0 & 0 & 1 \end{pmatrix} \qquad (k_{15} = 3)$$

$$= \tfrac{3}{2}\,A + \tfrac{1}{4}\,AB - \tfrac{1}{2}\,BA - \tfrac{1}{4}\,ABB - \tfrac{1}{4}\,BAB + \tfrac{1}{4}\,BABB$$

$$\rho(T_{16}) = \begin{pmatrix} -1 & 1 & -1 \\ -2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix} \qquad (k_{16} = 3)$$

$$= -\tfrac{3}{4}I - \tfrac{3}{4}\,A + B + \tfrac{1}{4}\,AB + \tfrac{1}{4}\,BA - \tfrac{1}{4}\,BB + \tfrac{1}{4}\,BAB$$

$$\rho(T_{17}) = \begin{pmatrix} 1 & -1 & 0 \\ 2 & 0 & -2 \\ 4 & 0 & -1 \end{pmatrix} = BA \qquad (k_{17} = 3)$$

$$\rho(T_{18}) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & -2 & 3 \end{pmatrix} \qquad (k_{18} = 3)$$

$$= \tfrac{3}{4}I - \tfrac{3}{4}\,A + \tfrac{1}{2}\,B + AB + \tfrac{1}{4}\,BA - \tfrac{1}{4}\,BB - \tfrac{1}{4}\,ABB - \tfrac{1}{2}\,BAB + \tfrac{1}{4}\,BABB$$

$$\rho(T_{19}) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 2 & -1 \end{pmatrix} \qquad (k_{19} = 3)$$

$$= -\tfrac{3}{4}I - \tfrac{3}{4}\,A - \tfrac{1}{2}\,B + \tfrac{1}{4}\,AB + \tfrac{1}{4}\,BA + \tfrac{1}{4}\,BB + \tfrac{1}{4}\,BAB$$

$$\rho(T_{20}) = \begin{pmatrix} -1 & -1 & 1 \\ -2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix} \qquad (k_{20} = 3)$$

$$= -\tfrac{3}{4}I - \tfrac{3}{4}\,A - \tfrac{1}{2}\,B - \tfrac{5}{4}\,AB + \tfrac{1}{4}\,BA + \tfrac{1}{4}\,BB + \tfrac{1}{2}\,ABB + \tfrac{1}{4}\,BAB \qquad .$$

Block 3: $M_2(\mathbb{C})$ on χ_6 (degree 8)

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} -\frac{1}{2} & 1 \\ -\frac{7}{4} & -\frac{1}{2} \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$A^2 - I = 0,$$
$$B^2 + B + 2I = 0,$$
$$BA = -A - AB,$$
$$ABA = -I - B,$$
$$ABB = -2\,A - AB.$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$
$$\rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A \qquad (k_2 = 1)$$
$$\rho(T_3) = \begin{pmatrix} -\frac{1}{2} & 1 \\ -\frac{7}{4} & -\frac{1}{2} \end{pmatrix} = B \qquad (k_3 = 3)$$
$$\rho(T_4) = \begin{pmatrix} -\frac{1}{2} & -1 \\ \frac{7}{4} & -\frac{1}{2} \end{pmatrix} = -I - B \qquad (k_4 = 3)$$
$$\rho(T_5) = \begin{pmatrix} -\frac{1}{2} & 1 \\ \frac{7}{4} & \frac{1}{2} \end{pmatrix} = AB \qquad (k_5 = 3)$$
$$\rho(T_6) = \begin{pmatrix} -\frac{1}{2} & -1 \\ -\frac{7}{4} & \frac{1}{2} \end{pmatrix} = -A - AB \qquad (k_6 = 3)$$
$$\rho(T_7) = \begin{pmatrix} -\frac{1}{2} & 1 \\ \frac{7}{4} & \frac{1}{2} \end{pmatrix} = AB \qquad (k_7 = 3)$$
$$\rho(T_8) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = -A \qquad (k_8 = 3)$$
$$\rho(T_9) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I \qquad (k_9 = 3)$$
$$\rho(T_{10}) = \begin{pmatrix} \frac{1}{2} & 1 \\ -\frac{7}{4} & -\frac{3}{2} \end{pmatrix} = A + B \qquad (k_{10} = 3)$$
$$\rho(T_{11}) = \begin{pmatrix} \frac{1}{2} & -1 \\ -\frac{7}{4} & \frac{3}{2} \end{pmatrix} = I - A - AB \qquad (k_{11} = 3)$$
$$\rho(T_{12}) = \begin{pmatrix} -\frac{1}{2} & -1 \\ \frac{7}{4} & -\frac{1}{2} \end{pmatrix} = -I - B \qquad (k_{12} = 3)$$
$$\rho(T_{13}) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} = I + A \qquad (k_{13} = 3)$$
$$\rho(T_{14}) = \begin{pmatrix} -\frac{1}{2} & 1 \\ -\frac{7}{4} & -\frac{1}{2} \end{pmatrix} = B \qquad (k_{14} = 3)$$
$$\rho(T_{15}) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = -A \qquad (k_{15} = 3)$$
$$\rho(T_{16}) = \begin{pmatrix} \frac{1}{2} & -1 \\ \frac{7}{4} & -\frac{3}{2} \end{pmatrix} = -I + A - B \qquad (k_{16} = 3)$$
$$\rho(T_{17}) = \begin{pmatrix} -\frac{1}{2} & -1 \\ -\frac{7}{4} & \frac{1}{2} \end{pmatrix} = -A - AB \qquad (k_{17} = 3)$$

$$\rho(T_{18}) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I \qquad (k_{18} = 3)$$

$$\rho(T_{19}) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} = I + A \qquad (k_{19} = 3)$$

$$\rho(T_{20}) = \begin{pmatrix} \frac{1}{2} & 1 \\ \frac{7}{4} & \frac{3}{2} \end{pmatrix} = I + AB \qquad (k_{20} = 3).$$

6 Klein $\{3,7\}$ graph: $\big(\mathrm{PGL}(2,7),\, D_7\big)$

Graph. The 1-skeleton of the $\{3,7\}$ tiling of the Klein quartic: 24 vertices, valency 7 (the dual of the $\{7,3\}$ map). $G = \mathrm{Aut}\,\Gamma \cong \mathrm{PGL}(2,7)$ ($|G| = 336$, GAP-verified), with vertex stabilizer $H \cong D_7$ (dihedral of order 14).

Algebra. Rank 6, $\mathcal{H} \cong \mathbb{C}^2 \oplus M_2(\mathbb{C})$.

Highlights. The stabilizer fixes three vertices, and the two non-trivial valency-1 orbitals act as the ω-diagonal pair ($A^2 + A + I = 0$, so $\rho(T_2)$ has order 3). The symmetric valency-7 orbital satisfies

$$B^2 = 7I$$

— the "why-N" constant equals the valency on the nose, the cleanest instance in the catalogue — and the chiral valency-7 pair is exactly $\rho(T_5) = BA$, $\rho(T_6) = AB$, with $AB + BA = -B$.

One-dimensional characters.

	$\chi_1\,(1)$	$\chi_7\,(7)$
T_1	1	1
T_2	1	1
T_3	1	1
T_4	7	-1
T_5	7	-1
T_6	7	-1

Block 1: $M_2(\mathbb{C})$ on χ_9 (degree 8)

Generators.

$$A = \rho(T_2) = \begin{pmatrix} \omega & 0 \\ 0 & \bar{\omega} \end{pmatrix}, \qquad B = \rho(T_4) = \begin{pmatrix} 0 & \sqrt{7} \\ \sqrt{7} & 0 \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$A^2 + A + I = 0,$$
$$B^2 - 7I = 0,$$
$$BA = -B - AB,$$
$$ABA = B,$$
$$ABB = 7\,A.$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$

$$\rho(T_2) = \begin{pmatrix} \omega & 0 \\ 0 & \bar{\omega} \end{pmatrix} = A \qquad (k_2 = 1)$$

$$\rho(T_3) = \begin{pmatrix} \bar{\omega} & 0 \\ 0 & \omega \end{pmatrix} = -I - A \qquad (k_3 = 1)$$

$$\rho(T_4) = \begin{pmatrix} 0 & \sqrt{7} \\ \sqrt{7} & 0 \end{pmatrix} = B \qquad (k_4 = 7)$$

$$\rho(T_5) = \begin{pmatrix} 0 & \bar{\omega}\sqrt{7} \\ \omega\sqrt{7} & 0 \end{pmatrix} = -B - AB \qquad (k_5 = 7)$$

$$\rho(T_6) = \begin{pmatrix} 0 & \omega\sqrt{7} \\ \bar{\omega}\sqrt{7} & 0 \end{pmatrix} = AB \qquad (k_6 = 7).$$

7 Klein $\{7,3\}$ graph: $\big(\mathrm{PGL}(2,7),\, S_3\big)$

Graph. The 1-skeleton of the $\{7,3\}$ tiling of the Klein quartic (56 vertices, cubic, girth 7): the *same* coset space $\mathrm{PSL}(2,7)/C_3$ as F056A, but a girth-7 orbital. The full group is $G = \mathrm{Aut}\,\Gamma \cong \mathrm{PGL}(2,7)$ ($|G| = 336$; GAP-verified, and visible in the character data: the degree-6 constituents take values in $\mathbb{Q}(\zeta_8)$, so G has elements of order 8 — ruling out $\mathrm{PSL}(2,7) \times C_2$), with vertex stabilizer $H \cong S_3$.

Algebra. Rank 13, $\mathcal{H} \cong \mathbb{C}^5 \oplus M_2(\mathbb{C}) \oplus M_2(\mathbb{C})$.

Highlights. Block 1 is integral with $B^2 + B + 4I = 0$ and the companion orbital relations producing eigenvalues $\frac{-1\pm\sqrt{17}}{2}$; Block 2 contains an orbital of order 3 and one with square $8I$.

One-dimensional characters.

	$\chi_1\,(1)$	$\chi_3\,(6)$	$\chi_4\,(6)$	$\chi_5\,(6)$	$\chi_6\,(7)$
T_1	1	1	1	1	1
T_2	1	-1	1	1	-1
T_3	3	2	$-\sqrt{2}$	$\sqrt{2}$	0
T_4	3	-2	$-\sqrt{2}$	$\sqrt{2}$	0
T_5	3	-2	$-\sqrt{2}$	$\sqrt{2}$	0
T_6	3	2	$-\sqrt{2}$	$\sqrt{2}$	0
T_7	6	-1	-1	-1	3
T_8	6	-1	-1	-1	3
T_9	6	0	$-2+2\sqrt{2}$	$-2-2\sqrt{2}$	0
T_{10}	6	-2	$2+\sqrt{2}$	$2-\sqrt{2}$	0
T_{11}	6	1	-1	-1	-3
T_{12}	6	2	$2+\sqrt{2}$	$2-\sqrt{2}$	0
T_{13}	6	1	-1	-1	-3

Block 1: $M_2(\mathbb{C})$ on χ_7 (degree 7)

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} -1 & 1 \\ -4 & 0 \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$A^2 - I = 0,$$
$$B^2 + B + 4I = 0,$$

$$BA = -I - A - AB,$$
$$ABA = -I - A - B,$$
$$ABB = -4\,A - AB.$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$
$$\rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A \qquad (k_2 = 1)$$
$$\rho(T_3) = \begin{pmatrix} -1 & 1 \\ -4 & 0 \end{pmatrix} = B \qquad (k_3 = 3)$$
$$\rho(T_4) = \begin{pmatrix} -1 & -1 \\ -4 & 0 \end{pmatrix} = -I - A - AB \qquad (k_4 = 3)$$
$$\rho(T_5) = \begin{pmatrix} -1 & 1 \\ 4 & 0 \end{pmatrix} = AB \qquad (k_5 = 3)$$
$$\rho(T_6) = \begin{pmatrix} -1 & -1 \\ 4 & 0 \end{pmatrix} = -I - A - B \qquad (k_6 = 3)$$
$$\rho(T_7) = \begin{pmatrix} 2 & 1 \\ -4 & -1 \end{pmatrix} = I + 2\,A + B \qquad (k_7 = 6)$$
$$\rho(T_8) = \begin{pmatrix} 2 & -1 \\ 4 & -1 \end{pmatrix} = A - B \qquad (k_8 = 6)$$
$$\rho(T_9) = \begin{pmatrix} -2 & 0 \\ 0 & 0 \end{pmatrix} = -I - A \qquad (k_9 = 6)$$
$$\rho(T_{10}) = \begin{pmatrix} -2 & 0 \\ 0 & 4 \end{pmatrix} = I - 3\,A \qquad (k_{10} = 6)$$
$$\rho(T_{11}) = \begin{pmatrix} 2 & -1 \\ -4 & 1 \end{pmatrix} = I - AB \qquad (k_{11} = 6)$$
$$\rho(T_{12}) = \begin{pmatrix} -2 & 0 \\ 0 & -4 \end{pmatrix} = -3I + A \qquad (k_{12} = 6)$$
$$\rho(T_{13}) = \begin{pmatrix} 2 & 1 \\ 4 & 1 \end{pmatrix} = 2I + A + AB \qquad (k_{13} = 6).$$

Block 2: $M_2(\mathbb{C})$ on χ_9 (degree 8)

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} \frac{1}{2} & 1 \\ -\frac{7}{4} & -\frac{3}{2} \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$A^2 - I = 0,$$
$$B^2 + B + I = 0,$$
$$BA = 2I - A - AB,$$
$$ABA = -I + 2\,A - B,$$
$$ABB = -A - AB.$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$

$$\rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A \qquad (k_2 = 1)$$

$$\rho(T_3) = \begin{pmatrix} \frac{1}{2} & 1 \\ -\frac{7}{4} & -\frac{3}{2} \end{pmatrix} = B \qquad (k_3 = 3)$$

$$\rho(T_4) = \begin{pmatrix} \frac{1}{2} & -1 \\ -\frac{7}{4} & \frac{3}{2} \end{pmatrix} = 2I - A - AB \qquad (k_4 = 3)$$

$$\rho(T_5) = \begin{pmatrix} \frac{1}{2} & 1 \\ \frac{7}{4} & \frac{3}{2} \end{pmatrix} = AB \qquad (k_5 = 3)$$

$$\rho(T_6) = \begin{pmatrix} \frac{1}{2} & -1 \\ \frac{7}{4} & -\frac{3}{2} \end{pmatrix} = -I + 2\,A - B \qquad (k_6 = 3)$$

$$\rho(T_7) = \begin{pmatrix} -1 & -2 \\ \frac{7}{2} & -1 \end{pmatrix} = -2I + 2\,A - 2\,B \qquad (k_7 = 6)$$

$$\rho(T_8) = \begin{pmatrix} -1 & 2 \\ -\frac{7}{2} & -1 \end{pmatrix} = -2\,A + 2\,B \qquad (k_8 = 6)$$

$$\rho(T_9) = \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix} = 2I + 2\,A \qquad (k_9 = 6)$$

$$\rho(T_{10}) = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix} = -2I \qquad (k_{10} = 6)$$

$$\rho(T_{11}) = \begin{pmatrix} -1 & 2 \\ \frac{7}{2} & 1 \end{pmatrix} = -2I + 2\,AB \qquad (k_{11} = 6)$$

$$\rho(T_{12}) = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix} = -2\,A \qquad (k_{12} = 6)$$

$$\rho(T_{13}) = \begin{pmatrix} -1 & -2 \\ -\frac{7}{2} & 1 \end{pmatrix} = 2I - 2\,A - 2\,AB \qquad (k_{13} = 6).$$

Remark 2. *The order-3 matrix* $\left(\begin{smallmatrix} 1/2 & 1 \\ -7/4 & -3/2 \end{smallmatrix}\right)$ *is* literally the same *as* $\rho(T_{10})$ *in F056A's Block 3 — same coset geometry, different orbital, identical* 2×2 *block.*

8 Zhou-3 (F182A): $\big(\mathrm{PSL}(2,13),\ S_3\big)$

Graph. The 182-vertex cubic coset graph of girth 7; $G = \mathrm{PSL}(2,13)$ ($|G| = 1092$), $H = S_3$.

Algebra. Rank 34, $\mathcal{H} \cong \mathbb{C}^4 \oplus M_2(\mathbb{C})^{\oplus 3} \oplus M_3(\mathbb{C})^{\oplus 2}$.

Highlights. The three M_2 blocks (degrees $12, 12, 12$) are Galois conjugate over $\mathbb{Q}(\zeta_7)^+$ via $c_k \mapsto c_{2k}$. Block 4 is the integral Steinberg M_3 with $B^3 = -I$ and an orbital of characteristic polynomial $x^3 + x - 26$, echoing $\mathrm{PSL}(2,7)$'s $x^3 + x - 6$. Block 5 carries $B^3 = 5I$ — the first *cubic* analogue of Meinhold's $B^2 = 13I$. The $\pm\sqrt{13}$ of $\mathbb{Q}(\zeta_{13})$ appears in the 1-dimensional characters (quadratic Gauss periods $\frac{-1\pm\sqrt{13}}{2}$), a different 13 from Meinhold's.

One-dimensional characters.

	$\chi_1\,(1)$	$\chi_2\,(7)$	$\chi_3\,(7)$	$\chi_8\,(14)$
T_1	1	1	1	1
T_2	1	-1	-1	-1
T_3	3	$\frac{1}{2}+\frac{1}{2}\sqrt{13}$	$\frac{1}{2}-\frac{1}{2}\sqrt{13}$	1
T_4	3	$-\frac{1}{2}-\frac{1}{2}\sqrt{13}$	$-\frac{1}{2}+\frac{1}{2}\sqrt{13}$	-1
T_5	3	$\frac{1}{2}+\frac{1}{2}\sqrt{13}$	$\frac{1}{2}-\frac{1}{2}\sqrt{13}$	1
T_6	3	$-\frac{1}{2}-\frac{1}{2}\sqrt{13}$	$-\frac{1}{2}+\frac{1}{2}\sqrt{13}$	-1
T_7	6	$\frac{1}{2}+\frac{1}{2}\sqrt{13}$	$\frac{1}{2}-\frac{1}{2}\sqrt{13}$	-2
T_8	6	0	0	0
T_9	6	$-\frac{1}{2}-\frac{1}{2}\sqrt{13}$	$-\frac{1}{2}+\frac{1}{2}\sqrt{13}$	2
T_{10}	6	$\frac{5}{2}-\frac{1}{2}\sqrt{13}$	$\frac{5}{2}+\frac{1}{2}\sqrt{13}$	-1
T_{11}	6	$-\frac{5}{2}+\frac{1}{2}\sqrt{13}$	$-\frac{5}{2}-\frac{1}{2}\sqrt{13}$	1
T_{12}	6	0	0	0
T_{13}	6	$-\frac{1}{2}-\frac{1}{2}\sqrt{13}$	$-\frac{1}{2}+\frac{1}{2}\sqrt{13}$	2
T_{14}	6	$\frac{1}{2}+\frac{1}{2}\sqrt{13}$	$\frac{1}{2}-\frac{1}{2}\sqrt{13}$	-2
T_{15}	6	$-\frac{5}{2}+\frac{1}{2}\sqrt{13}$	$-\frac{5}{2}-\frac{1}{2}\sqrt{13}$	1
T_{16}	6	$\frac{5}{2}-\frac{1}{2}\sqrt{13}$	$\frac{5}{2}+\frac{1}{2}\sqrt{13}$	-1
T_{17}	6	$-\frac{1}{2}-\frac{1}{2}\sqrt{13}$	$-\frac{1}{2}+\frac{1}{2}\sqrt{13}$	2
T_{18}	6	$\frac{1}{2}+\frac{1}{2}\sqrt{13}$	$\frac{1}{2}-\frac{1}{2}\sqrt{13}$	-2
T_{19}	6	$\frac{1}{2}+\frac{1}{2}\sqrt{13}$	$\frac{1}{2}-\frac{1}{2}\sqrt{13}$	-2
T_{20}	6	$-\frac{1}{2}-\frac{1}{2}\sqrt{13}$	$-\frac{1}{2}+\frac{1}{2}\sqrt{13}$	2
T_{21}	6	3	3	0
T_{22}	6	-3	-3	0
T_{23}	6	0	0	0
T_{24}	6	$\frac{1}{2}+\frac{1}{2}\sqrt{13}$	$\frac{1}{2}-\frac{1}{2}\sqrt{13}$	1
T_{25}	6	$-\frac{1}{2}-\frac{1}{2}\sqrt{13}$	$-\frac{1}{2}+\frac{1}{2}\sqrt{13}$	-1
T_{26}	6	3	3	0
T_{27}	6	0	0	3
T_{28}	6	0	0	-3
T_{29}	6	0	0	0
T_{30}	6	$\frac{1}{2}+\frac{1}{2}\sqrt{13}$	$\frac{1}{2}-\frac{1}{2}\sqrt{13}$	1
T_{31}	6	$-\frac{1}{2}-\frac{1}{2}\sqrt{13}$	$-\frac{1}{2}+\frac{1}{2}\sqrt{13}$	-1
T_{32}	6	0	0	0
T_{33}	6	-3	-3	0
T_{34}	6	0	0	0

Block 1: $M_2(\mathbb{C})$ on χ_4 (degree 12)

Generators.

$$A=\rho(T_2)=\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix},\qquad B=\rho(T_3)=\begin{pmatrix}-\frac{1}{2}+\frac{1}{2}c_3 & 1\\ -\frac{11}{4}-\frac{5}{4}c_1-c_2 & -\frac{1}{2}c_1+\frac{1}{2}c_2\end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$A^2-I=0,$$
$$B^2+\big(1+c_1\big)B+\big(3+2c_1+c_2\big)I=0,$$

$$
\begin{aligned}
BA &= \big(-1-c_2\big)I + \big(-1-c_1\big)A - AB,\\
ABA &= \big(-1-c_1\big)I + \big(-1-c_2\big)A - B,\\
ABB &= \big(-3-2c_1-c_2\big)A + \big(-1-c_1\big)AB.
\end{aligned}
$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$
\begin{aligned}
\rho(T_1) &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I && (k_1 = 1)\\
\rho(T_2) &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A && (k_2 = 1)\\
\rho(T_3) &= \begin{pmatrix} -\frac{1}{2} + \frac{1}{2}c_3 & 1 \\ -\frac{11}{4} - \frac{5}{4}c_1 - c_2 & -\frac{1}{2}c_1 + \frac{1}{2}c_2 \end{pmatrix} = B && (k_3 = 3)\\
\rho(T_4) &= \begin{pmatrix} -\frac{1}{2} + \frac{1}{2}c_3 & -1 \\ -\frac{11}{4} - \frac{5}{4}c_1 - c_2 & \frac{1}{2}c_1 - \frac{1}{2}c_2 \end{pmatrix} && (k_4 = 3)\\
&= \big(-1-c_2\big)I + \big(-1-c_1\big)A - AB\\
\rho(T_5) &= \begin{pmatrix} -\frac{1}{2} + \frac{1}{2}c_3 & -1 \\ \frac{11}{4} + \frac{5}{4}c_1 + c_2 & -\frac{1}{2}c_1 + \frac{1}{2}c_2 \end{pmatrix} && (k_5 = 3)\\
&= \big(-1-c_1\big)I + \big(-1-c_2\big)A - B\\
\rho(T_6) &= \begin{pmatrix} -\frac{1}{2} + \frac{1}{2}c_3 & 1 \\ \frac{11}{4} + \frac{5}{4}c_1 + c_2 & \frac{1}{2}c_1 - \frac{1}{2}c_2 \end{pmatrix} = AB && (k_6 = 3)\\
\rho(T_7) &= \begin{pmatrix} -\frac{1}{2}c_1 + c_2 & c_2 \\ -\frac{1}{4} - c_1 + \frac{1}{2}c_2 & \frac{1}{2}c_1 + c_3 \end{pmatrix} = -c_1 I + c_2\, A + c_2\, AB && (k_7 = 6)\\
\rho(T_8) &= \begin{pmatrix} -2c_2 & 0 \\ 0 & 0 \end{pmatrix} = -c_2 I - c_2\, A && (k_8 = 6)\\
\rho(T_9) &= \begin{pmatrix} -\frac{1}{2}c_1 + c_2 & -c_2 \\ -\frac{1}{4} - c_1 + \frac{1}{2}c_2 & -\frac{1}{2}c_1 - c_3 \end{pmatrix} = \big(1+c_2\big)I - A - c_2\, B && (k_9 = 6)\\
\rho(T_{10}) &= \begin{pmatrix} -\frac{1}{2} & -c_1 - 2c_2 \\ 1 + \frac{3}{4}c_1 + \frac{5}{4}c_2 & -\frac{1}{2} - c_3 \end{pmatrix} = -A + \big(-c_1 - 2c_2\big)B && (k_{10} = 6)\\
\rho(T_{11}) &= \begin{pmatrix} -\frac{1}{2} & c_1 + 2c_2 \\ 1 + \frac{3}{4}c_1 + \frac{5}{4}c_2 & \frac{1}{2} + c_3 \end{pmatrix} && (k_{11} = 6)\\
&= \big(-c_1 - c_2\big)I + \big(c_1 + c_2\big)A + \big(c_1 + 2c_2\big)AB\\
\rho(T_{12}) &= \begin{pmatrix} -2c_2 & 0 \\ 0 & 0 \end{pmatrix} = -c_2 I - c_2\, A && (k_{12} = 6)\\
\rho(T_{13}) &= \begin{pmatrix} -\frac{1}{2}c_1 + c_2 & c_2 \\ \frac{1}{4} + c_1 - \frac{1}{2}c_2 & -\frac{1}{2}c_1 - c_3 \end{pmatrix} = c_2 I - c_1\, A + c_2\, B && (k_{13} = 6)\\
\rho(T_{14}) &= \begin{pmatrix} -\frac{1}{2}c_1 + c_2 & -c_2 \\ \frac{1}{4} + c_1 - \frac{1}{2}c_2 & \frac{1}{2}c_1 + c_3 \end{pmatrix} = -I + \big(1+c_2\big)A - c_2\, AB && (k_{14} = 6)\\
\rho(T_{15}) &= \begin{pmatrix} -\frac{1}{2} & -c_1 - 2c_2 \\ -1 - \frac{3}{4}c_1 - \frac{5}{4}c_2 & \frac{1}{2} + c_3 \end{pmatrix} = -I + \big(-c_1 - 2c_2\big)AB && (k_{15} = 6)\\
\rho(T_{16}) &= \begin{pmatrix} -\frac{1}{2} & c_1 + 2c_2 \\ -1 - \frac{3}{4}c_1 - \frac{5}{4}c_2 & -\frac{1}{2} - c_3 \end{pmatrix} && (k_{16} = 6)\\
&= \big(c_1 + c_2\big)I + \big(-c_1 - c_2\big)A + \big(c_1 + 2c_2\big)B
\end{aligned}
$$

$$\rho(T_{17}) = \begin{pmatrix} 1+2c_1+\frac{3}{2}c_2 & -1-c_2 \\ \frac{5}{2}+\frac{1}{4}c_1+\frac{3}{2}c_2 & -\frac{1}{2}c_2+c_3 \end{pmatrix} \qquad (k_{17}=6)$$
$$= \big(2c_1+c_2\big)A + \big(-1-c_2\big)B$$

$$\rho(T_{18}) = \begin{pmatrix} 1+2c_1+\frac{3}{2}c_2 & 1+c_2 \\ \frac{5}{2}+\frac{1}{4}c_1+\frac{3}{2}c_2 & \frac{1}{2}c_2-c_3 \end{pmatrix} \qquad (k_{18}=6)$$
$$= \big(-c_1-2c_3\big)I + c_1\,A + \big(1+c_2\big)AB$$

$$\rho(T_{19}) = \begin{pmatrix} 1+2c_1+\frac{3}{2}c_2 & -1-c_2 \\ -\frac{5}{2}-\frac{1}{4}c_1-\frac{3}{2}c_2 & \frac{1}{2}c_2-c_3 \end{pmatrix} \qquad (k_{19}=6)$$
$$= \big(2c_1+c_2\big)I + \big(-1-c_2\big)AB$$

$$\rho(T_{20}) = \begin{pmatrix} 1+2c_1+\frac{3}{2}c_2 & 1+c_2 \\ -\frac{5}{2}-\frac{1}{4}c_1-\frac{3}{2}c_2 & -\frac{1}{2}c_2+c_3 \end{pmatrix} \qquad (k_{20}=6)$$
$$= c_1 I + \big(-c_1-2c_3\big)A + \big(1+c_2\big)B$$

$$\rho(T_{21}) = \begin{pmatrix} -\frac{1}{2} & -c_1 \\ \frac{3}{2}+\frac{11}{4}c_1+\frac{1}{4}c_2 & -\frac{1}{2}+c_3 \end{pmatrix} \qquad (k_{21}=6)$$
$$= \big(-1+c_3\big)I + \big(1+c_2\big)A - c_1\,B$$

$$\rho(T_{22}) = \begin{pmatrix} -\frac{1}{2} & c_1 \\ \frac{3}{2}+\frac{11}{4}c_1+\frac{1}{4}c_2 & \frac{1}{2}-c_3 \end{pmatrix} = c_1 I + c_1\,AB \qquad (k_{22}=6)$$

$$\rho(T_{23}) = \begin{pmatrix} 1-c_1 & 2c_1+2c_2 \\ 0 & 0 \end{pmatrix} = I + A + \big(c_1+c_2\big)B + \big(c_1+c_2\big)AB \qquad (k_{23}=6)$$

$$\rho(T_{24}) = \begin{pmatrix} -c_2-\frac{1}{2}c_3 & 1-c_1-2c_2 \\ -\frac{7}{4}-\frac{1}{2}c_1+\frac{1}{4}c_2 & \frac{1}{2}+\frac{3}{2}c_1-\frac{1}{2}c_2 \end{pmatrix} \qquad (k_{24}=6)$$
$$= \big(-2c_2-c_3\big)I + c_2\,A + \big(1-c_1-2c_2\big)B$$

$$\rho(T_{25}) = \begin{pmatrix} -c_2-\frac{1}{2}c_3 & -1+c_1+2c_2 \\ -\frac{7}{4}-\frac{1}{2}c_1+\frac{1}{4}c_2 & -\frac{1}{2}-\frac{3}{2}c_1+\frac{1}{2}c_2 \end{pmatrix} \qquad (k_{25}=6)$$
$$= \big(-c_1-c_2\big)I + c_1\,A + \big(-1+c_1+2c_2\big)AB$$

$$\rho(T_{26}) = \begin{pmatrix} -\frac{1}{2} & c_1 \\ -\frac{3}{2}-\frac{11}{4}c_1-\frac{1}{4}c_2 & -\frac{1}{2}+c_3 \end{pmatrix} = c_1\,A + c_1\,B \qquad (k_{26}=6)$$

$$\rho(T_{27}) = \begin{pmatrix} -1+c_3 & 0 \\ 0 & -3-c_3 \end{pmatrix} = -2I + \big(-c_1-c_2\big)A \qquad (k_{27}=6)$$

$$\rho(T_{28}) = \begin{pmatrix} -1+c_3 & 0 \\ 0 & 3+c_3 \end{pmatrix} = \big(-c_1-c_2\big)I - 2\,A \qquad (k_{28}=6)$$

$$\rho(T_{29}) = \begin{pmatrix} 1-c_1 & 0 \\ -\frac{5}{2}-\frac{7}{2}c_1-\frac{3}{2}c_2 & 0 \end{pmatrix} \qquad (k_{29}=6)$$
$$= \big(1+c_2\big)I + \big(-c_1-c_2\big)A + \big(c_1+c_2\big)B + \big(-c_1-c_2\big)AB$$

$$\rho(T_{30}) = \begin{pmatrix} -c_2-\frac{1}{2}c_3 & -1+c_1+2c_2 \\ \frac{7}{4}+\frac{1}{2}c_1-\frac{1}{4}c_2 & \frac{1}{2}+\frac{3}{2}c_1-\frac{1}{2}c_2 \end{pmatrix} \qquad (k_{30}=6)$$
$$= c_1 I + \big(-c_1-c_2\big)A + \big(-1+c_1+2c_2\big)B$$

$$\rho(T_{31}) = \begin{pmatrix} -c_2-\frac{1}{2}c_3 & 1-c_1-2c_2 \\ \frac{7}{4}+\frac{1}{2}c_1-\frac{1}{4}c_2 & -\frac{1}{2}-\frac{3}{2}c_1+\frac{1}{2}c_2 \end{pmatrix} \qquad (k_{31}=6)$$
$$= c_2 I + \big(-2c_2-c_3\big)A + \big(1-c_1-2c_2\big)AB$$

$$\rho(T_{32}) = \begin{pmatrix} 1-c_1 & -2c_1-2c_2 \\ 0 & 0 \end{pmatrix} \qquad (k_{32}=6)$$
$$= -c_1 I - c_1\,A + \big(-c_1-c_2\big)B + \big(-c_1-c_2\big)AB$$
$$\rho(T_{33}) = \begin{pmatrix} -\frac{1}{2} & -c_1 \\ -\frac{3}{2}-\frac{11}{4}c_1-\frac{1}{4}c_2 & \frac{1}{2}-c_3 \end{pmatrix} \qquad (k_{33}=6)$$
$$= \big(1+c_2\big)I + \big(-1+c_3\big)A - c_1\,AB$$
$$\rho(T_{34}) = \begin{pmatrix} 1-c_1 & 0 \\ \frac{5}{2}+\frac{7}{2}c_1+\frac{3}{2}c_2 & 0 \end{pmatrix} \qquad (k_{34}=6)$$
$$= \big(-c_1-c_2\big)I + \big(1+c_2\big)A + \big(-c_1-c_2\big)B + \big(c_1+c_2\big)AB \qquad .$$

Block 2: $M_2(\mathbb{C})$ on χ_5 (degree 12)

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} -\frac{1}{2}+\frac{1}{2}c_2 & 1 \\ -\frac{3}{2}+\frac{1}{4}c_1+\frac{5}{4}c_2 & \frac{1}{2}c_1-\frac{1}{2}c_3 \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$A^2 - I = 0,$$
$$B^2 + \big(-c_1-c_2\big)B + \big(1-c_1-2c_2\big)I = 0,$$
$$BA = \big(-1-c_1\big)I + \big(c_1+c_2\big)A - AB,$$
$$ABA = \big(c_1+c_2\big)I + \big(-1-c_1\big)A - B,$$
$$ABB = \big(-1+c_1+2c_2\big)A + \big(c_1+c_2\big)AB.$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \qquad (k_1=1)$$
$$\rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A \qquad (k_2=1)$$
$$\rho(T_3) = \begin{pmatrix} -\frac{1}{2}+\frac{1}{2}c_2 & 1 \\ -\frac{3}{2}+\frac{1}{4}c_1+\frac{5}{4}c_2 & \frac{1}{2}c_1-\frac{1}{2}c_3 \end{pmatrix} = B \qquad (k_3=3)$$
$$\rho(T_4) = \begin{pmatrix} -\frac{1}{2}+\frac{1}{2}c_2 & -1 \\ -\frac{3}{2}+\frac{1}{4}c_1+\frac{5}{4}c_2 & -\frac{1}{2}c_1+\frac{1}{2}c_3 \end{pmatrix} \qquad (k_4=3)$$
$$= \big(-1-c_1\big)I + \big(c_1+c_2\big)A - AB$$
$$\rho(T_5) = \begin{pmatrix} -\frac{1}{2}+\frac{1}{2}c_2 & -1 \\ \frac{3}{2}-\frac{1}{4}c_1-\frac{5}{4}c_2 & \frac{1}{2}c_1-\frac{1}{2}c_3 \end{pmatrix} \qquad (k_5=3)$$
$$= \big(c_1+c_2\big)I + \big(-1-c_1\big)A - B$$
$$\rho(T_6) = \begin{pmatrix} -\frac{1}{2}+\frac{1}{2}c_2 & 1 \\ \frac{3}{2}-\frac{1}{4}c_1-\frac{5}{4}c_2 & -\frac{1}{2}c_1+\frac{1}{2}c_3 \end{pmatrix} = AB \qquad (k_6=3)$$
$$\rho(T_7) = \begin{pmatrix} c_1-\frac{1}{2}c_3 & c_1 \\ \frac{3}{4}+\frac{3}{2}c_1+c_2 & c_2+\frac{1}{2}c_3 \end{pmatrix} = -c_3 I + c_1\,A + c_1\,AB \qquad (k_7=6)$$
$$\rho(T_8) = \begin{pmatrix} -2c_1 & 0 \\ 0 & 0 \end{pmatrix} = -c_1 I - c_1\,A \qquad (k_8=6)$$

$$\rho(T_9) = \begin{pmatrix} c_1 - \frac{1}{2}c_3 & -c_1 \\ \frac{3}{4} + \frac{3}{2}c_1 + c_2 & -c_2 - \frac{1}{2}c_3 \end{pmatrix} = \big(1 + c_1\big)I - A - c_1\, B \qquad (k_9 = 6)$$

$$\rho(T_{10}) = \begin{pmatrix} -\frac{1}{2} & -2c_1 - c_3 \\ \frac{1}{4} + \frac{1}{2}c_1 - \frac{3}{4}c_2 & -\frac{1}{2} - c_2 \end{pmatrix} = -A + \big(-2c_1 - c_3\big)B \qquad (k_{10} = 6)$$

$$\rho(T_{11}) = \begin{pmatrix} -\frac{1}{2} & 2c_1 + c_3 \\ \frac{1}{4} + \frac{1}{2}c_1 - \frac{3}{4}c_2 & \frac{1}{2} + c_2 \end{pmatrix} \qquad (k_{11} = 6)$$
$$= \big(1 + c_2\big)I + \big(-1 - c_2\big)A + \big(2c_1 + c_3\big)AB$$

$$\rho(T_{12}) = \begin{pmatrix} -2c_1 & 0 \\ 0 & 0 \end{pmatrix} = -c_1 I - c_1\, A \qquad (k_{12} = 6)$$

$$\rho(T_{13}) = \begin{pmatrix} c_1 - \frac{1}{2}c_3 & c_1 \\ -\frac{3}{4} - \frac{3}{2}c_1 - c_2 & -c_2 - \frac{1}{2}c_3 \end{pmatrix} = c_1 I - c_3\, A + c_1\, B \qquad (k_{13} = 6)$$

$$\rho(T_{14}) = \begin{pmatrix} c_1 - \frac{1}{2}c_3 & -c_1 \\ -\frac{3}{4} - \frac{3}{2}c_1 - c_2 & c_2 + \frac{1}{2}c_3 \end{pmatrix} = -I + \big(1 + c_1\big)A - c_1\, AB \qquad (k_{14} = 6)$$

$$\rho(T_{15}) = \begin{pmatrix} -\frac{1}{2} & -2c_1 - c_3 \\ -\frac{1}{4} - \frac{1}{2}c_1 + \frac{3}{4}c_2 & \frac{1}{2} + c_2 \end{pmatrix} = -I + \big(-2c_1 - c_3\big)AB \qquad (k_{15} = 6)$$

$$\rho(T_{16}) = \begin{pmatrix} -\frac{1}{2} & 2c_1 + c_3 \\ -\frac{1}{4} - \frac{1}{2}c_1 + \frac{3}{4}c_2 & -\frac{1}{2} - c_2 \end{pmatrix} \qquad (k_{16} = 6)$$
$$= \big(-1 - c_2\big)I + \big(1 + c_2\big)A + \big(2c_1 + c_3\big)B$$

$$\rho(T_{17}) = \begin{pmatrix} -1 - \frac{1}{2}c_1 - 2c_2 & -1 - c_1 \\ \frac{9}{4} + \frac{5}{4}c_1 - \frac{1}{4}c_2 & -\frac{1}{2}c_1 + c_2 \end{pmatrix} \qquad (k_{17} = 6)$$
$$= \big(c_1 + 2c_3\big)A + \big(-1 - c_1\big)B$$

$$\rho(T_{18}) = \begin{pmatrix} -1 - \frac{1}{2}c_1 - 2c_2 & 1 + c_1 \\ \frac{9}{4} + \frac{5}{4}c_1 - \frac{1}{4}c_2 & \frac{1}{2}c_1 - c_2 \end{pmatrix} \qquad (k_{18} = 6)$$
$$= \big(-2c_2 - c_3\big)I + c_3\, A + \big(1 + c_1\big)AB$$

$$\rho(T_{19}) = \begin{pmatrix} -1 - \frac{1}{2}c_1 - 2c_2 & -1 - c_1 \\ -\frac{9}{4} - \frac{5}{4}c_1 + \frac{1}{4}c_2 & \frac{1}{2}c_1 - c_2 \end{pmatrix} \qquad (k_{19} = 6)$$
$$= \big(c_1 + 2c_3\big)I + \big(-1 - c_1\big)AB$$

$$\rho(T_{20}) = \begin{pmatrix} -1 - \frac{1}{2}c_1 - 2c_2 & 1 + c_1 \\ -\frac{9}{4} - \frac{5}{4}c_1 + \frac{1}{4}c_2 & -\frac{1}{2}c_1 + c_2 \end{pmatrix} \qquad (k_{20} = 6)$$
$$= c_3 I + \big(-2c_2 - c_3\big)A + \big(1 + c_1\big)B$$

$$\rho(T_{21}) = \begin{pmatrix} -\frac{1}{2} & -c_3 \\ -\frac{5}{4} - \frac{5}{2}c_1 - \frac{11}{4}c_2 & -\frac{1}{2} + c_2 \end{pmatrix} \qquad (k_{21} = 6)$$
$$= \big(-1 + c_2\big)I + \big(1 + c_1\big)A - c_3\, B$$

$$\rho(T_{22}) = \begin{pmatrix} -\frac{1}{2} & c_3 \\ -\frac{5}{4} - \frac{5}{2}c_1 - \frac{11}{4}c_2 & \frac{1}{2} - c_2 \end{pmatrix} = c_3 I + c_3\, AB \qquad (k_{22} = 6)$$

$$\rho(T_{23}) = \begin{pmatrix} 1 - c_3 & -2 - 2c_2 \\ 0 & 0 \end{pmatrix} = I + A + \big(-1 - c_2\big)B + \big(-1 - c_2\big)AB \qquad (k_{23} = 6)$$

$$\rho(T_{24}) = \begin{pmatrix} -c_1 - \frac{1}{2}c_2 & 2 - c_1 + c_2 \\ -\frac{5}{4} + \frac{3}{4}c_1 + \frac{1}{2}c_2 & -1 - 2c_1 - \frac{3}{2}c_2 \end{pmatrix} \qquad (k_{24} = 6)$$
$$= \big(-2c_1 - c_2\big)I + c_1\, A + \big(2 - c_1 + c_2\big)B$$

$$\rho(T_{25}) = \begin{pmatrix} -c_1 - \frac{1}{2}c_2 & -2 + c_1 - c_2 \\ -\frac{5}{4} + \frac{3}{4}c_1 + \frac{1}{2}c_2 & 1 + 2c_1 + \frac{3}{2}c_2 \end{pmatrix} \qquad (k_{25} = 6)$$
$$= \big(1 + c_2\big)I + c_3\, A + \big(-2 + c_1 - c_2\big)AB$$
$$\rho(T_{26}) = \begin{pmatrix} -\frac{1}{2} & c_3 \\ \frac{5}{4} + \frac{5}{2}c_1 + \frac{11}{4}c_2 & -\frac{1}{2} + c_2 \end{pmatrix} = c_3\, A + c_3\, B \qquad (k_{26} = 6)$$
$$\rho(T_{27}) = \begin{pmatrix} -1 + c_2 & 0 \\ 0 & -3 - c_2 \end{pmatrix} = -2I + \big(1 + c_2\big)A \qquad (k_{27} = 6)$$
$$\rho(T_{28}) = \begin{pmatrix} -1 + c_2 & 0 \\ 0 & 3 + c_2 \end{pmatrix} = \big(1 + c_2\big)I - 2\, A \qquad (k_{28} = 6)$$
$$\rho(T_{29}) = \begin{pmatrix} 1 - c_3 & 0 \\ 1 + 2c_1 + \frac{7}{2}c_2 & 0 \end{pmatrix} \qquad (k_{29} = 6)$$
$$= \big(1 + c_1\big)I + \big(1 + c_2\big)A + \big(-1 - c_2\big)B + \big(1 + c_2\big)AB$$
$$\rho(T_{30}) = \begin{pmatrix} -c_1 - \frac{1}{2}c_2 & -2 + c_1 - c_2 \\ \frac{5}{4} - \frac{3}{4}c_1 - \frac{1}{2}c_2 & -1 - 2c_1 - \frac{3}{2}c_2 \end{pmatrix} \qquad (k_{30} = 6)$$
$$= c_3 I + \big(1 + c_2\big)A + \big(-2 + c_1 - c_2\big)B$$
$$\rho(T_{31}) = \begin{pmatrix} -c_1 - \frac{1}{2}c_2 & 2 - c_1 + c_2 \\ \frac{5}{4} - \frac{3}{4}c_1 - \frac{1}{2}c_2 & 1 + 2c_1 + \frac{3}{2}c_2 \end{pmatrix} \qquad (k_{31} = 6)$$
$$= c_1 I + \big(-2c_1 - c_2\big)A + \big(2 - c_1 + c_2\big)AB$$
$$\rho(T_{32}) = \begin{pmatrix} 1 - c_3 & 2 + 2c_2 \\ 0 & 0 \end{pmatrix} = -c_3 I - c_3\, A + \big(1 + c_2\big)B + \big(1 + c_2\big)AB \qquad (k_{32} = 6)$$
$$\rho(T_{33}) = \begin{pmatrix} -\frac{1}{2} & -c_3 \\ \frac{5}{4} + \frac{5}{2}c_1 + \frac{11}{4}c_2 & \frac{1}{2} - c_2 \end{pmatrix} \qquad (k_{33} = 6)$$
$$= \big(1 + c_1\big)I + \big(-1 + c_2\big)A - c_3\, AB$$
$$\rho(T_{34}) = \begin{pmatrix} 1 - c_3 & 0 \\ -1 - 2c_1 - \frac{7}{2}c_2 & 0 \end{pmatrix} \qquad (k_{34} = 6)$$
$$= \big(1 + c_2\big)I + \big(1 + c_1\big)A + \big(1 + c_2\big)B + \big(-1 - c_2\big)AB \qquad .$$

Block 3: $M_2(\mathbb{C})$ **on** χ_6 **(degree 12)**

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} -\frac{1}{2} + \frac{1}{2}c_1 & 1 \\ -\frac{7}{4} + c_1 - \frac{1}{4}c_2 & -\frac{1}{2}c_2 + \frac{1}{2}c_3 \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$A^2 - I = 0,$$
$$B^2 + \big(1 + c_2\big)B + \big(2 - c_1 + c_2\big)I = 0,$$
$$BA = \big(c_1 + c_2\big)I + \big(-1 - c_2\big)A - AB,$$
$$ABA = \big(-1 - c_2\big)I + \big(c_1 + c_2\big)A - B,$$
$$ABB = \big(-2 + c_1 - c_2\big)A + \big(-1 - c_2\big)AB.$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$

$$\rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A \qquad (k_2 = 1)$$

$$\rho(T_3) = \begin{pmatrix} -\frac{1}{2} + \frac{1}{2}c_1 & 1 \\ -\frac{7}{4} + c_1 - \frac{1}{4}c_2 & -\frac{1}{2}c_2 + \frac{1}{2}c_3 \end{pmatrix} = B \qquad (k_3 = 3)$$

$$\rho(T_4) = \begin{pmatrix} -\frac{1}{2} + \frac{1}{2}c_1 & -1 \\ -\frac{7}{4} + c_1 - \frac{1}{4}c_2 & \frac{1}{2}c_2 - \frac{1}{2}c_3 \end{pmatrix} \qquad (k_4 = 3)$$
$$= \big(c_1 + c_2\big)I + \big(-1 - c_2\big)A - AB$$

$$\rho(T_5) = \begin{pmatrix} -\frac{1}{2} + \frac{1}{2}c_1 & -1 \\ \frac{7}{4} - c_1 + \frac{1}{4}c_2 & -\frac{1}{2}c_2 + \frac{1}{2}c_3 \end{pmatrix} \qquad (k_5 = 3)$$
$$= \big(-1 - c_2\big)I + \big(c_1 + c_2\big)A - B$$

$$\rho(T_6) = \begin{pmatrix} -\frac{1}{2} + \frac{1}{2}c_1 & 1 \\ \frac{7}{4} - c_1 + \frac{1}{4}c_2 & \frac{1}{2}c_2 - \frac{1}{2}c_3 \end{pmatrix} = AB \qquad (k_6 = 3)$$

$$\rho(T_7) = \begin{pmatrix} -\frac{1}{2}c_2 + c_3 & c_3 \\ -\frac{3}{4} - \frac{1}{2}c_1 - \frac{3}{2}c_2 & c_1 + \frac{1}{2}c_2 \end{pmatrix} = -c_2 I + c_3\, A + c_3\, AB \qquad (k_7 = 6)$$

$$\rho(T_8) = \begin{pmatrix} -2c_3 & 0 \\ 0 & 0 \end{pmatrix} = -c_3 I - c_3\, A \qquad (k_8 = 6)$$

$$\rho(T_9) = \begin{pmatrix} -\frac{1}{2}c_2 + c_3 & -c_3 \\ -\frac{3}{4} - \frac{1}{2}c_1 - \frac{3}{2}c_2 & -c_1 - \frac{1}{2}c_2 \end{pmatrix} \qquad (k_9 = 6)$$
$$= \big(-c_1 - c_2\big)I - A - c_3\, B$$

$$\rho(T_{10}) = \begin{pmatrix} -\frac{1}{2} & -c_2 - 2c_3 \\ -\frac{1}{4} - \frac{5}{4}c_1 - \frac{1}{2}c_2 & -\frac{1}{2} - c_1 \end{pmatrix} = -A + \big(-c_2 - 2c_3\big)B \qquad (k_{10} = 6)$$

$$\rho(T_{11}) = \begin{pmatrix} -\frac{1}{2} & c_2 + 2c_3 \\ -\frac{1}{4} - \frac{5}{4}c_1 - \frac{1}{2}c_2 & \frac{1}{2} + c_1 \end{pmatrix} \qquad (k_{11} = 6)$$
$$= \big(1 + c_1\big)I + \big(-1 - c_1\big)A + \big(c_2 + 2c_3\big)AB$$

$$\rho(T_{12}) = \begin{pmatrix} -2c_3 & 0 \\ 0 & 0 \end{pmatrix} = -c_3 I - c_3\, A \qquad (k_{12} = 6)$$

$$\rho(T_{13}) = \begin{pmatrix} -\frac{1}{2}c_2 + c_3 & c_3 \\ \frac{3}{4} + \frac{1}{2}c_1 + \frac{3}{2}c_2 & -c_1 - \frac{1}{2}c_2 \end{pmatrix} = c_3 I - c_2\, A + c_3\, B \qquad (k_{13} = 6)$$

$$\rho(T_{14}) = \begin{pmatrix} -\frac{1}{2}c_2 + c_3 & -c_3 \\ \frac{3}{4} + \frac{1}{2}c_1 + \frac{3}{2}c_2 & c_1 + \frac{1}{2}c_2 \end{pmatrix} \qquad (k_{14} = 6)$$
$$= -I + \big(-c_1 - c_2\big)A - c_3\, AB$$

$$\rho(T_{15}) = \begin{pmatrix} -\frac{1}{2} & -c_2 - 2c_3 \\ \frac{1}{4} + \frac{5}{4}c_1 + \frac{1}{2}c_2 & \frac{1}{2} + c_1 \end{pmatrix} = -I + \big(-c_2 - 2c_3\big)AB \qquad (k_{15} = 6)$$

$$\rho(T_{16}) = \begin{pmatrix} -\frac{1}{2} & c_2 + 2c_3 \\ \frac{1}{4} + \frac{5}{4}c_1 + \frac{1}{2}c_2 & -\frac{1}{2} - c_1 \end{pmatrix} \qquad (k_{16} = 6)$$
$$= \big(-1 - c_1\big)I + \big(1 + c_1\big)A + \big(c_2 + 2c_3\big)B$$

$$\rho(T_{17}) = \begin{pmatrix} -\frac{1}{2} - \frac{3}{2}c_1 + \frac{1}{2}c_2 & c_1 + c_2 \\ 1 - \frac{3}{2}c_1 - \frac{5}{4}c_2 & c_1 - \frac{1}{2}c_3 \end{pmatrix} \qquad (k_{17} = 6)$$
$$= \big(2c_2 + c_3\big)A + \big(c_1 + c_2\big)B$$

$$\rho(T_{18}) = \begin{pmatrix} -\frac{1}{2} - \frac{3}{2}c_1 + \frac{1}{2}c_2 & -c_1 - c_2 \\ 1 - \frac{3}{2}c_1 - \frac{5}{4}c_2 & -c_1 + \frac{1}{2}c_3 \end{pmatrix} \qquad (k_{18} = 6)$$

$$= \big(-2c_1 - c_2\big)I + c_2\,A + \big(-c_1 - c_2\big)AB$$

$$\rho(T_{19}) = \begin{pmatrix} -\frac{1}{2} - \frac{3}{2}c_1 + \frac{1}{2}c_2 & c_1 + c_2 \\ -1 + \frac{3}{2}c_1 + \frac{5}{4}c_2 & -c_1 + \frac{1}{2}c_3 \end{pmatrix} \qquad (k_{19} = 6)$$

$$= \big(2c_2 + c_3\big)I + \big(c_1 + c_2\big)AB$$

$$\rho(T_{20}) = \begin{pmatrix} -\frac{1}{2} - \frac{3}{2}c_1 + \frac{1}{2}c_2 & -c_1 - c_2 \\ -1 + \frac{3}{2}c_1 + \frac{5}{4}c_2 & c_1 - \frac{1}{2}c_3 \end{pmatrix} \qquad (k_{20} = 6)$$

$$= c_2 I + \big(-2c_1 - c_2\big)A + \big(-c_1 - c_2\big)B$$

$$\rho(T_{21}) = \begin{pmatrix} -\frac{1}{2} & -c_2 \\ \frac{5}{4} - \frac{1}{4}c_1 + \frac{5}{2}c_2 & -\frac{1}{2} + c_1 \end{pmatrix} \qquad (k_{21} = 6)$$

$$= \big(-1 + c_1\big)I + \big(-c_1 - c_2\big)A - c_2\,B$$

$$\rho(T_{22}) = \begin{pmatrix} -\frac{1}{2} & c_2 \\ \frac{5}{4} - \frac{1}{4}c_1 + \frac{5}{2}c_2 & \frac{1}{2} - c_1 \end{pmatrix} = c_2 I + c_2\,AB \qquad (k_{22} = 6)$$

$$\rho(T_{23}) = \begin{pmatrix} 1 - c_2 & -2 - 2c_1 \\ 0 & 0 \end{pmatrix} = I + A + \big(-1 - c_1\big)B + \big(-1 - c_1\big)AB \qquad (k_{23} = 6)$$

$$\rho(T_{24}) = \begin{pmatrix} -\frac{1}{2}c_1 - c_3 & 3 + 2c_1 + c_2 \\ -2 - \frac{1}{4}c_1 - \frac{3}{4}c_2 & 1 + \frac{1}{2}c_1 + 2c_2 \end{pmatrix} \qquad (k_{24} = 6)$$

$$= \big(-c_1 - 2c_3\big)I + c_3\,A + \big(3 + 2c_1 + c_2\big)B$$

$$\rho(T_{25}) = \begin{pmatrix} -\frac{1}{2}c_1 - c_3 & -3 - 2c_1 - c_2 \\ -2 - \frac{1}{4}c_1 - \frac{3}{4}c_2 & -1 - \frac{1}{2}c_1 - 2c_2 \end{pmatrix} \qquad (k_{25} = 6)$$

$$= \big(1 + c_1\big)I + c_2\,A + \big(-3 - 2c_1 - c_2\big)AB$$

$$\rho(T_{26}) = \begin{pmatrix} -\frac{1}{2} & c_2 \\ -\frac{5}{4} + \frac{1}{4}c_1 - \frac{5}{2}c_2 & -\frac{1}{2} + c_1 \end{pmatrix} = c_2\,A + c_2\,B \qquad (k_{26} = 6)$$

$$\rho(T_{27}) = \begin{pmatrix} -1 + c_1 & 0 \\ 0 & -3 - c_1 \end{pmatrix} = -2I + \big(1 + c_1\big)A \qquad (k_{27} = 6)$$

$$\rho(T_{28}) = \begin{pmatrix} -1 + c_1 & 0 \\ 0 & 3 + c_1 \end{pmatrix} = \big(1 + c_1\big)I - 2\,A \qquad (k_{28} = 6)$$

$$\rho(T_{29}) = \begin{pmatrix} 1 - c_2 & 0 \\ -1 + \frac{3}{2}c_1 - 2c_2 & 0 \end{pmatrix} \qquad (k_{29} = 6)$$

$$= \big(-c_1 - c_2\big)I + \big(1 + c_1\big)A + \big(-1 - c_1\big)B + \big(1 + c_1\big)AB$$

$$\rho(T_{30}) = \begin{pmatrix} -\frac{1}{2}c_1 - c_3 & -3 - 2c_1 - c_2 \\ 2 + \frac{1}{4}c_1 + \frac{3}{4}c_2 & 1 + \frac{1}{2}c_1 + 2c_2 \end{pmatrix} \qquad (k_{30} = 6)$$

$$= c_2 I + \big(1 + c_1\big)A + \big(-3 - 2c_1 - c_2\big)B$$

$$\rho(T_{31}) = \begin{pmatrix} -\frac{1}{2}c_1 - c_3 & 3 + 2c_1 + c_2 \\ 2 + \frac{1}{4}c_1 + \frac{3}{4}c_2 & -1 - \frac{1}{2}c_1 - 2c_2 \end{pmatrix} \qquad (k_{31} = 6)$$

$$= c_3 I + \big(-c_1 - 2c_3\big)A + \big(3 + 2c_1 + c_2\big)AB$$

$$\rho(T_{32}) = \begin{pmatrix} 1 - c_2 & 2 + 2c_1 \\ 0 & 0 \end{pmatrix} = -c_2 I - c_2\,A + \big(1 + c_1\big)B + \big(1 + c_1\big)AB \qquad (k_{32} = 6)$$

$$\rho(T_{33}) = \begin{pmatrix} -\frac{1}{2} & -c_2 \\ -\frac{5}{4} + \frac{1}{4}c_1 - \frac{5}{2}c_2 & \frac{1}{2} - c_1 \end{pmatrix} \qquad (k_{33} = 6)$$

$$= \big(-c_1 - c_2\big)I + \big(-1 + c_1\big)A - c_2\,AB$$

$$\rho(T_{34}) = \begin{pmatrix} 1-c_2 & 0 \\ 1-\frac{3}{2}c_1+2c_2 & 0 \end{pmatrix} \qquad (k_{34}=6)$$
$$= \big(1+c_1\big)I + \big(-c_1-c_2\big)A + \big(1+c_1\big)B + \big(-1-c_1\big)AB \qquad .$$

Block 4: $M_3(\mathbb{C})$ on χ_7 (degree 13)

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} 2 & -1 & -2 \\ 2 & -2 & -3 \\ 1 & 0 & 0 \end{pmatrix}.$$

Complete relations (monomial basis I, A, B, AB, BA, BB, ABB, BAB, $BABB$ of $M_3(\mathbb{C})$):

$$A^2 - I = 0,$$
$$B^3 + I = 0,$$
$$ABA = -B + AB + BA,$$
$$BAA = B,$$
$$BBA = 4I + BB - ABB - BAB,$$
$$BBB = -I,$$
$$ABBA = 4\,A - BAB,$$
$$ABBB = -A,$$
$$BABA = 4I - ABB,$$
$$BABBA = I - A - 4\,B + 4\,BA + BABB,$$
$$BABBB = -BA.$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$

$$\rho(T_2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = A \qquad (k_2 = 1)$$

$$\rho(T_3) = \begin{pmatrix} 2 & -1 & -2 \\ 2 & -2 & -3 \\ 1 & 0 & 0 \end{pmatrix} = B \qquad (k_3 = 3)$$

$$\rho(T_4) = \begin{pmatrix} 2 & -1 & 2 \\ 2 & -2 & 3 \\ 1 & 0 & 0 \end{pmatrix} = BA \qquad (k_4 = 3)$$

$$\rho(T_5) = \begin{pmatrix} 2 & -1 & 2 \\ 2 & -2 & 3 \\ -1 & 0 & 0 \end{pmatrix} = -B + AB + BA \qquad (k_5 = 3)$$

$$\rho(T_6) = \begin{pmatrix} 2 & -1 & -2 \\ 2 & -2 & -3 \\ -1 & 0 & 0 \end{pmatrix} = AB \qquad (k_6 = 3)$$

$$\rho(T_7) = \begin{pmatrix} 1 & 0 & 3 \\ 3 & -1 & 8 \\ 0 & 1 & 3 \end{pmatrix} = -I - 2\,A - AB + BA - BB + ABB + BAB \qquad (k_7 = 6)$$

$$\rho(T_8) = \begin{pmatrix} 6 & -2 & 0 \\ 14 & -4 & 0 \\ 0 & 0 & 0 \end{pmatrix} = -A - 2\,B - AB + BA - BB + BAB + BABB \qquad (k_8 = 6)$$

$$\rho(T_9) = \begin{pmatrix} 1 & 0 & -3 \\ 3 & -1 & -8 \\ 0 & 1 & -3 \end{pmatrix} = -2I + 3\,A + 2\,B - AB - BA - BB \qquad (k_9 = 6)$$

$$\rho(T_{10}) = \begin{pmatrix} 1 & -1 & -2 \\ -4 & 1 & -3 \\ 1 & 0 & 1 \end{pmatrix} = 2I + A + B + BB - BAB \qquad (k_{10} = 6)$$

$$\rho(T_{11}) = \begin{pmatrix} 1 & -1 & 2 \\ -4 & 1 & 3 \\ 1 & 0 & -1 \end{pmatrix} = I + 2\,A + BA + BB - BAB \qquad (k_{11} = 6)$$

$$\rho(T_{12}) = \begin{pmatrix} 0 & 2 & 0 \\ -2 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} = -I + 2\,B + AB - BA - BB + BAB - BABB \qquad (k_{12} = 6)$$

$$\rho(T_{13}) = \begin{pmatrix} 1 & 0 & 3 \\ 3 & -1 & 8 \\ 0 & -1 & -3 \end{pmatrix} = -2I - A - 2\,B + AB + BA + BAB \qquad (k_{13} = 6)$$

$$\rho(T_{14}) = \begin{pmatrix} 1 & 0 & -3 \\ 3 & -1 & -8 \\ 0 & -1 & 3 \end{pmatrix} = 3I - 2\,A + AB - BA - ABB \qquad (k_{14} = 6)$$

$$\rho(T_{15}) = \begin{pmatrix} 1 & -1 & -2 \\ -4 & 1 & -3 \\ -1 & 0 & -1 \end{pmatrix} = I + 2\,A + AB + BB - BAB \qquad (k_{15} = 6)$$

$$\rho(T_{16}) = \begin{pmatrix} 1 & -1 & 2 \\ -4 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix} = 2I + A - B + AB + BA + BB - BAB \qquad (k_{16} = 6)$$

$$\rho(T_{17}) = \begin{pmatrix} 1 & 0 & -1 \\ 3 & -1 & 2 \\ 2 & -1 & 1 \end{pmatrix} = -3\,A + BAB \qquad (k_{17} = 6)$$

$$\rho(T_{18}) = \begin{pmatrix} 1 & 0 & 1 \\ 3 & -1 & -2 \\ 2 & -1 & -1 \end{pmatrix} = I - ABB \qquad (k_{18} = 6)$$

$$\rho(T_{19}) = \begin{pmatrix} 1 & 0 & -1 \\ 3 & -1 & 2 \\ -2 & 1 & -1 \end{pmatrix} = -3I - BB + ABB + BAB \qquad (k_{19} = 6)$$

$$\rho(T_{20}) = \begin{pmatrix} 1 & 0 & 1 \\ 3 & -1 & -2 \\ -2 & 1 & 1 \end{pmatrix} = A - BB \qquad (k_{20} = 6)$$

$$\rho(T_{21}) = \begin{pmatrix} -2 & 1 & -3 \\ -5 & 4 & -1 \\ 3 & -1 & -2 \end{pmatrix} = B - AB - BA + BB \qquad (k_{21} = 6)$$

$$\rho(T_{22}) = \begin{pmatrix} -2 & 1 & 3 \\ -5 & 4 & 1 \\ 3 & -1 & 2 \end{pmatrix} = 4I - AB + BB - ABB - BAB \qquad (k_{22} = 6)$$

$$\rho(T_{23}) = \begin{pmatrix} 2 & -2 & -2 \\ 8 & -4 & 4 \\ 0 & 0 & 0 \end{pmatrix} \qquad (k_{23} = 6)$$

$$= -2I - 3\,A - 2\,B - AB + BA + ABB + BAB + BABB$$

$$\rho(T_{24}) = \begin{pmatrix} -3 & 0 & 3 \\ -3 & -1 & 8 \\ 0 & -1 & -1 \end{pmatrix} = -I - 2\,A - 2\,B + AB + BA + BB \qquad (k_{24} = 6)$$

$$\rho(T_{25}) = \begin{pmatrix} -3 & 0 & -3 \\ -3 & -1 & -8 \\ 0 & -1 & 1 \end{pmatrix} = 2I - A + AB - BA + BB - ABB - BAB \qquad (k_{25} = 6)$$

$$\rho(T_{26}) = \begin{pmatrix} -2 & 1 & 3 \\ -5 & 4 & 1 \\ -3 & 1 & -2 \end{pmatrix} = 4\,A - B - BAB \qquad (k_{26} = 6)$$

$$\rho(T_{27}) = \begin{pmatrix} -2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = -2\,A - AB - BA - BB + BAB \qquad (k_{27} = 6)$$

$$\rho(T_{28}) = \begin{pmatrix} -2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & -2 \end{pmatrix} = -2I - AB - BA - BB + BAB \qquad (k_{28} = 6)$$

$$\rho(T_{29}) = \begin{pmatrix} -4 & 2 & 0 \\ -8 & 2 & 0 \\ 4 & -2 & 0 \end{pmatrix} = I - 2\,A + 2\,B + AB - BA + BB - ABB - BABB \qquad (k_{29} = 6)$$

$$\rho(T_{30}) = \begin{pmatrix} -3 & 0 & -3 \\ -3 & -1 & -8 \\ 0 & 1 & -1 \end{pmatrix} = -I + 2\,A + 2\,B - AB - BA - BAB \qquad (k_{30} = 6)$$

$$\rho(T_{31}) = \begin{pmatrix} -3 & 0 & 3 \\ -3 & -1 & 8 \\ 0 & 1 & 1 \end{pmatrix} = -2I - A - AB + BA + ABB \qquad (k_{31} = 6)$$

$$\rho(T_{32}) = \begin{pmatrix} 2 & -2 & 2 \\ 8 & -4 & -4 \\ 0 & 0 & 0 \end{pmatrix} = 2I + A - 2\,B - AB + BA - ABB - BAB + BABB \qquad (k_{32} = 6)$$

$$\rho(T_{33}) = \begin{pmatrix} -2 & 1 & -3 \\ -5 & 4 & -1 \\ -3 & 1 & 2 \end{pmatrix} = -BA + ABB \qquad (k_{33} = 6)$$

$$\rho(T_{34}) = \begin{pmatrix} -4 & 2 & 0 \\ -8 & 2 & 0 \\ -4 & 2 & 0 \end{pmatrix} = -3I + 2\,A + 2\,B + AB - BA - BB + ABB - BABB \qquad (k_{34} = 6).$$

Block 5: $M_3(\mathbb{C})$ **on** χ_9 **(degree 14)**

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{11}{4} \\ 2 & 1 & 3 \\ 1 & 0 & -\frac{3}{2} \end{pmatrix}.$$

Complete relations (monomial basis I, A, B, AB, BA, BB, ABB, BAB, $BABB$ of $M_3(\mathbb{C})$):

$$\begin{gathered} A^2 - I = 0, \\ B^3 - 5I = 0, \\ ABA = -3I + 3\,A - B + AB + BA, \\ BAA = B, \\ BBA = I + 3\,B + BB - ABB - BAB, \\ BBB = 5I, \\ ABBA = A + 3\,B - BAB, \\ ABBB = 5\,A, \\ BABA = I + 3\,BA - ABB, \\ BABBA = -5I + 5\,A - B + BA + BABB, \\ BABBB = 5\,BA. \end{gathered}$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$

$$\rho(T_2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = A \qquad (k_2 = 1)$$

$$\rho(T_3) = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{11}{4} \\ 2 & 1 & 3 \\ 1 & 0 & -\frac{3}{2} \end{pmatrix} = B \qquad (k_3 = 3)$$

$$\rho(T_4) = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & \frac{11}{4} \\ 2 & 1 & -3 \\ 1 & 0 & \frac{3}{2} \end{pmatrix} = BA \qquad (k_4 = 3)$$

$$\rho(T_5) = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & \frac{11}{4} \\ 2 & 1 & -3 \\ -1 & 0 & -\frac{3}{2} \end{pmatrix} = -3I + 3\,A - B + AB + BA \qquad (k_5 = 3)$$

$$\rho(T_6) = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{11}{4} \\ 2 & 1 & 3 \\ -1 & 0 & \frac{3}{2} \end{pmatrix} = AB \qquad (k_6 = 3)$$

$$\begin{aligned} \rho(T_7) &= \begin{pmatrix} -2 & -\frac{3}{2} & -3 \\ 0 & 2 & 8 \\ 0 & 1 & 0 \end{pmatrix} \qquad (k_7 = 6) \\ &= 8I - 2\,A + 6\,B - 4\,AB - 2\,BA + 2\,BB - 2\,ABB - 2\,BAB \end{aligned}$$

$$\rho(T_8) = \begin{pmatrix} -6 & -\frac{7}{2} & 0 \\ 8 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \qquad (k_8 = 6)$$
$$= 9I - 7\,A + 7\,B - AB - 2\,BA + 5\,BB - 3\,ABB - 2\,BAB - 2\,BABB$$

$$\rho(T_9) = \begin{pmatrix} -2 & -\frac{3}{2} & 3 \\ 0 & 2 & -8 \\ 0 & 1 & 0 \end{pmatrix} = 10I - 6\,A + 2\,B - 4\,AB - 4\,BA + 2\,BB \qquad (k_9 = 6)$$

$$\rho(T_{10}) = \begin{pmatrix} 1 & -\frac{1}{4} & 1 \\ -4 & -2 & 6 \\ -2 & -\frac{3}{2} & -2 \end{pmatrix} \qquad (k_{10} = 6)$$
$$= -10I + 10\,A - 2\,B + 3\,AB + 3\,BA - 2\,BB - BAB$$

$$\rho(T_{11}) = \begin{pmatrix} 1 & -\frac{1}{4} & -1 \\ -4 & -2 & -6 \\ -2 & -\frac{3}{2} & 2 \end{pmatrix} \qquad (k_{11} = 6)$$
$$= -2I - A - 6\,B + 3\,AB - 2\,BA - 2\,BB + 3\,ABB + 2\,BAB$$

$$\rho(T_{12}) = \begin{pmatrix} 0 & 2 & 0 \\ -8 & -4 & 0 \\ 0 & 0 & 0 \end{pmatrix} \qquad (k_{12} = 6)$$
$$= -4I + 3\,A - 4\,B - 2\,AB - BA - 4\,BB + 3\,ABB + BAB + 2\,BABB$$

$$\rho(T_{13}) = \begin{pmatrix} -2 & -\frac{3}{2} & -3 \\ 0 & 2 & 8 \\ 0 & -1 & 0 \end{pmatrix} = 4I + 2\,A + 4\,B - 2\,AB - 2\,BA - 2\,BAB \qquad (k_{13} = 6)$$

$$\rho(T_{14}) = \begin{pmatrix} -2 & -\frac{3}{2} & 3 \\ 0 & 2 & -8 \\ 0 & -1 & 0 \end{pmatrix} = 6I - 2\,A - 2\,AB - 4\,BA + 2\,ABB \qquad (k_{14} = 6)$$

$$\rho(T_{15}) = \begin{pmatrix} 1 & -\frac{1}{4} & 1 \\ -4 & -2 & 6 \\ 2 & \frac{3}{2} & 2 \end{pmatrix} = I - A + 3\,B - 2\,AB + 3\,BA + BB - 3\,ABB - BAB \qquad (k_{15} = 6)$$

$$\rho(T_{16}) = \begin{pmatrix} 1 & -\frac{1}{4} & -1 \\ -4 & -2 & -6 \\ 2 & \frac{3}{2} & -2 \end{pmatrix} = 5I - 8\,A - B - 2\,AB - 2\,BA + BB + 2\,BAB \qquad (k_{16} = 6)$$

$$\rho(T_{17}) = \begin{pmatrix} 1 & \frac{3}{4} & -4 \\ 0 & -1 & 2 \\ 2 & \frac{1}{2} & -2 \end{pmatrix} = -3\,A + BAB \qquad (k_{17} = 6)$$

$$\rho(T_{18}) = \begin{pmatrix} 1 & \frac{3}{4} & 4 \\ 0 & -1 & -2 \\ 2 & \frac{1}{2} & 2 \end{pmatrix} = -2I + 3\,BA - ABB \qquad (k_{18} = 6)$$

$$\rho(T_{19}) = \begin{pmatrix} 1 & \frac{3}{4} & -4 \\ 0 & -1 & 2 \\ -2 & -\frac{1}{2} & 2 \end{pmatrix} = -3I - 3\,B + 3\,AB - BB + ABB + BAB \qquad (k_{19} = 6)$$

$$\rho(T_{20}) = \begin{pmatrix} 1 & \frac{3}{4} & 4 \\ 0 & -1 & -2 \\ -2 & -\frac{1}{2} & -2 \end{pmatrix} = -9I + 7\,A - 3\,B + 3\,AB + 3\,BA - BB \qquad (k_{20} = 6)$$

$$\rho(T_{21}) = \begin{pmatrix} -2 & \frac{1}{4} & \frac{3}{2} \\ 4 & 1 & -4 \\ 0 & \frac{1}{2} & 1 \end{pmatrix} = 3I - 3\,A + B - AB - BA + BB \qquad (k_{21} = 6)$$

$$\rho(T_{22}) = \begin{pmatrix} -2 & \frac{1}{4} & -\frac{3}{2} \\ 4 & 1 & 4 \\ 0 & \frac{1}{2} & -1 \end{pmatrix} = I + 3\,B - AB + BB - ABB - BAB \qquad (k_{22} = 6)$$

$$\rho(T_{23}) = \begin{pmatrix} 2 & \frac{5}{2} & 4 \\ -4 & -4 & -2 \\ 0 & 0 & 0 \end{pmatrix} \qquad (k_{23} = 6)$$

$$= -11I + 6\,A - 5\,B + 2\,AB + 4\,BA - 3\,BB + ABB + BAB + BABB$$

$$\rho(T_{24}) = \begin{pmatrix} 0 & -\frac{3}{4} & \frac{3}{2} \\ 0 & 2 & -4 \\ 0 & \frac{1}{2} & -1 \end{pmatrix} = 5I - 2\,A + B - 2\,AB - 2\,BA + BB \qquad (k_{24} = 6)$$

$$\rho(T_{25}) = \begin{pmatrix} 0 & -\frac{3}{4} & -\frac{3}{2} \\ 0 & 2 & 4 \\ 0 & \frac{1}{2} & 1 \end{pmatrix} = 5I - A + 3\,B - 2\,AB - BA + BB - ABB - BAB \qquad (k_{25} = 6)$$

$$\rho(T_{26}) = \begin{pmatrix} -2 & \frac{1}{4} & -\frac{3}{2} \\ 4 & 1 & 4 \\ 0 & -\frac{1}{2} & 1 \end{pmatrix} = A + 2\,B - BAB \qquad (k_{26} = 6)$$

$$\rho(T_{27}) = \begin{pmatrix} 4 & \frac{1}{2} & 0 \\ -4 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix} = -3I + A - 3\,B + 2\,AB + 2\,BA - BB + BAB \qquad (k_{27} = 6)$$

$$\rho(T_{28}) = \begin{pmatrix} 4 & \frac{1}{2} & 0 \\ -4 & -1 & 0 \\ 0 & 0 & -2 \end{pmatrix} = -5I + 3\,A - 3\,B + 2\,AB + 2\,BA - BB + BAB \qquad (k_{28} = 6)$$

$$\rho(T_{29}) = \begin{pmatrix} -1 & -\frac{1}{4} & 0 \\ 4 & -1 & 0 \\ -2 & -\frac{1}{2} & 0 \end{pmatrix} \qquad (k_{29} = 6)$$

$$= -5I + A - B + 4\,AB + 2\,BA + BB - ABB - BABB$$

$$\rho(T_{30}) = \begin{pmatrix} 0 & -\frac{3}{4} & -\frac{3}{2} \\ 0 & 2 & 4 \\ 0 & -\frac{1}{2} & -1 \end{pmatrix} = 2I + 2\,A + 2\,B - AB - BA - BAB \qquad (k_{30} = 6)$$

$$\rho(T_{31}) = \begin{pmatrix} 0 & -\frac{3}{4} & \frac{3}{2} \\ 0 & 2 & -4 \\ 0 & -\frac{1}{2} & 1 \end{pmatrix} = 4I - A - AB - 2\,BA + ABB \qquad (k_{31} = 6)$$

$$\rho(T_{32}) = \begin{pmatrix} 2 & \frac{5}{2} & -4 \\ -4 & -4 & 2 \\ 0 & 0 & 0 \end{pmatrix} \qquad (k_{32} = 6)$$

$$= -7I + A - 5\,B + 2\,AB + BA - 3\,BB + 2\,ABB + 2\,BAB + BABB$$

$$\rho(T_{33}) = \begin{pmatrix} -2 & \frac{1}{4} & \frac{3}{2} \\ 4 & 1 & -4 \\ 0 & -\frac{1}{2} & -1 \end{pmatrix} = -BA + ABB \qquad (k_{33} = 6)$$

$$\rho(T_{34}) = \begin{pmatrix} -1 & -\frac{1}{4} & 0 \\ 4 & -1 & 0 \\ 2 & \frac{1}{2} & 0 \end{pmatrix} \qquad (k_{34} = 6)$$
$$= -4\,A + 2\,B + AB + 2\,BA + 2\,BB - 2\,ABB - BABB \qquad .$$

9 Meinhold-72: $\big(G_2(2),\ \mathrm{PSL}(3,2)\big)$

Graph. The Meinhold-72 orbital graph: $G = G_2(2) = \mathrm{PSU}(3,3){:}2$ ($|G| = 12096$) acting on the 72 cosets of $H = \mathrm{PSL}(3,2)$ (the $36 + 36$ chiral Fano structures).

Algebra. Rank 8, $\mathcal{H} \cong \mathbb{C}^2 \oplus M_2(\mathbb{C}) \oplus \mathbb{C}^2$.

Highlights. The block is generated by the mirror involution $A = \rho(T_2)$ and the directed valency-7 orbital $B = \rho(T_3)$ with

$$A^2 = I, \qquad B^2 + 4B + 13I = 0;$$

the roots $-2 \pm 3i$ of $x^2 + 4x + 13$ are Gaussian integers of norm 13. Equivalently, the symmetric generator $\rho(T_4) = -4A - AB$ satisfies $\rho(T_4)^2 = 13I$ and $A\rho(T_4) + \rho(T_4)A = -4I$. Why 13? $13 = 7{+}2{\cdot}3$: $NN^{\mathsf{T}} = 7I{+}2A_{\mathrm{SRG}}$ for the $\mathrm{SRG}(36, 21, 12, 12)$ on the chiral Fano structures, and $3 = \sqrt{k-\mu}$; the -4 is the $U_3(3)$-graph eigenvalue. (Full account: `why13.pdf`.)

One-dimensional characters.

	$\chi_1\,(1)$	$\chi_2\,(1)$	$\chi_{10}\,(21)$	$\chi_{11}\,(21)$
T_1	1	1	1	1
T_2	1	−1	−1	1
T_3	7	7	1	1
T_4	7	−7	−1	1
T_5	7	7	1	1
T_6	7	−7	−1	1
T_7	21	−21	3	−3
T_8	21	21	−3	−3

Block 1: $M_2(\mathbb{C})$ on χ_9 (degree 14)

Generators.

$$A = \rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \rho(T_3) = \begin{pmatrix} -2 & 1 \\ -9 & -2 \end{pmatrix}.$$

Complete relations (monomial basis $I,\ A,\ B,\ AB$ of $M_2(\mathbb{C})$):

$$A^2 - I = 0,$$
$$B^2 + 4\,B + 13I = 0,$$
$$BA = -4\,A - AB,$$
$$ABA = -4I - B,$$
$$ABB = -13\,A - 4\,AB.$$

All orbital images ($\rho(T_i)$, valency k_i, word in the generators):

$$\rho(T_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \qquad (k_1 = 1)$$

$$\rho(T_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A \qquad (k_2 = 1)$$

$$\rho(T_3) = \begin{pmatrix} -2 & 1 \\ -9 & -2 \end{pmatrix} = B \qquad (k_3 = 7)$$

$$\rho(T_4) = \begin{pmatrix} -2 & -1 \\ -9 & 2 \end{pmatrix} = -4\,A - AB \qquad (k_4 = 7)$$

$$\rho(T_5) = \begin{pmatrix} -2 & -1 \\ 9 & -2 \end{pmatrix} = -4I - B \qquad (k_5 = 7)$$

$$\rho(T_6) = \begin{pmatrix} -2 & 1 \\ 9 & 2 \end{pmatrix} = AB \qquad (k_6 = 7)$$

$$\rho(T_7) = \begin{pmatrix} 3 & 0 \\ 0 & -3 \end{pmatrix} = 3\,A \qquad (k_7 = 21)$$

$$\rho(T_8) = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = 3I \qquad (k_8 = 21).$$

Provenance

All data blocks: GAP, exact cyclotomic arithmetic, by isotypic restriction and spin-up of eigenvectors — single script `hecke_catalog_full.g` in `~/git/hecke/`, console audit trail `hecke_catalog_full.out`, LaTeX fragments `catalog-data/*.tex` (machine-generated; do not edit). Methods and proofs: `hecke-irreps.pdf`; the Meinhold story: `why13.pdf`. The non-Gelfand pairs $\mathrm{PSL}(2,19)/S_3$ and $\mathrm{PSL}(2,31)/S_4$ are deliberately out of scope.